



Parameter sensitivity analysis and calibration of a discrete element model for optimizing all-solid-state-battery cathode microstructures

Ali Mohammad Bazzoun^{a,b}, Javid Piruzjam^{a,c}, Steffen Hink^a, Lukas Rubacek^a, Alexander Fill^b, Thomas Carraro^c, Kai Peter Birke^b

^a Mercedes-Benz AG, Mercedesstraße 120, Stuttgart, 70372, Germany

^b Institute for Photovoltaics, University of Stuttgart, Pfaffenwaldring 47, Stuttgart, 70569, Germany

^c Applied Mathematics, Institute for Modelling and Computational Science, Faculty of Mechanical and Civil Engineering, Helmut Schmidt University/University of the Federal Armed Forces Hamburg, Holstenhofweg 85, Hamburg, 22043, Germany

ARTICLE INFO

Keywords:

All-solid-state-battery
Discrete element method
Cathode composite
Microstructure modeling
Sensitivity analysis

ABSTRACT

This study develops and evaluates a Discrete Element Method (DEM) model designed to simulate the produced cathode composite microstructures during uniaxial cold-pressing of All-Solid-State-Battery (ASSB) electrodes. The research primarily investigates how variations in model input parameters – such as particle size distribution, interparticle friction, and mechanical properties – affect critical performance indicators, including cathode active material (CAM) utilization (θ_{CAM}), packing behavior, porosity (ϵ), and solid electrolyte (SE) utilization (θ_{SE}), contributing to network formation and ionic percolation within DEM-generated composite cathodes. A comprehensive parameter sensitivity analysis highlights that frictional interactions are the most influential factors governing particle connectivity, percolation, and porosity, especially at higher CAM loadings (f_{CAM}). In addition to simulations, experimental measurements were performed to validate the DEM model and further analyze the cathode behavior under varying conditions. Both experimental and simulation results consistently demonstrate that increasing uniaxial pressure enhances discharge capacity by improving particle connectivity. However, while the DEM model exhibits high accuracy at lower f_{CAM} , it shows notable discrepancies and increased sensitivity to parameter variations at higher loadings, indicating the need for further refinements to achieve better alignment with experimental observations. This research provides essential insights into optimizing ASSB cathode designs, offering a robust framework for refining DEM models to predict and enhance battery performance under various conditions.

1. Introduction

Lithium-ion batteries (LIBs) are nearing their theoretical limits in energy and power density, prompting the exploration of advanced energy storage technologies [1]. All-solid-state batteries (ASSBs) have emerged as a promising successor to LIBs, offering the potential for significantly higher energy and power densities, along with enhanced safety features, making them particularly well-suited for automotive applications [1]. Among the various solid electrolytes (SEs) employed in ASSBs, sulfide-based SEs stand out as particularly suitable due to their high lithium-ion conductivities, which rival those of traditional liquid electrolytes [2]. Additionally, these sulfide-based SEs possess favorable mechanical properties for processing, such as softness and ductility, combined with a relatively low Young's modulus, further reinforcing their applicability in ASSB systems [3,4].

For the anode, high-capacity materials like lithium metal are the most promising for high energy-density ASSBs, given their high theoretical capacity of 3860 mAh g⁻¹ [5]. Alternative anodes, such as Ag-C composites without excess lithium [6], are also being explored as replacements for the currently most used graphite anode in LIBs, which has a theoretical capacity of 372 mAh g⁻¹ [7]. While the SE-Separator and anode are undoubtedly important, the cathode is often considered the decisive component, or bottleneck, for achieving high energy density in ASSB applications [8]. Current cathodes are typically composed of composites, including cathode active materials (CAM) like LiNi_xMn_yCo_{1-x-y}O₂ (NMC-type), SEs, and carbon additives (CA) [9], such as carbon black (CB) and carbon nanofibers (CNF) [10]. While binder-free fabrication of ASSB cathodes is possible at the laboratory

* Corresponding author at: Mercedes-Benz AG, Mercedesstraße 120, Stuttgart, 70372, Germany.

E-mail address: ali_mohammad.bazzoun@mercedes-benz.com (A.M. Bazzoun).

¹ These authors contributed equally to this paper and should be considered as co-first authors.

scale, scaling up to industrially relevant large cells requires converting these powder components into sheet-type intermediates before assembly and compression. Solvent-based synthesis routes have shown effectiveness in improving discharge capacity, capacity efficiency, and cycle retention in cathodes with high CAM content (89 wt%) [11]. However, the use of solvents introduces environmental and financial challenges, such as the need for drying processes and solvent recovery equipment. As an alternative, dry-film (DF) composite cathodes incorporating Poly(tetrafluoroethylene) (PTFE) binders have been developed and investigated for industrial-scale manufacturing, which can be densified through simple uniaxial compression [12].

Despite these advancements, cathode composites still face several challenges, including managing charge transport bottlenecks due to poor microstructural design and high porosity, addressing chemo-mechanical volume changes and interfacial stability issues, and balancing high CAM loading (f_{CAM}) with effective ionic and electronic conductivity [13]. These challenges are interrelated, often requiring a careful and balanced optimization of the cathode microstructure. This involves ensuring proper contact between CAM, SE, and CA, optimizing binder choice and amount, controlling particle size distributions, and managing the balance between different components to minimize porosity through improved microstructural composition and processing conditions [8]. One of the critical challenges in optimizing cathode composites is the issue of percolation, which directly impacts the battery's energy density and overall performance [14,15]. Effective ionic conduction relies on a well-maintained percolation network formed by the SE. However, as the fraction of CAM increases to achieve higher energy densities, the volume fraction of SE typically decreases, leading to a breakdown of the percolation network necessary for ionic conduction. This reduction results in lower utilization of the CAM, as ions struggle to reach all areas of the cathode [14]. To address this, the particle size ratio ($\lambda = \bar{D}_{\text{CAM}}/\bar{D}_{\text{SE}}$), where $\bar{D}_{\text{CAM}}$ and $\bar{D}_{\text{SE}}$ are the average diameters of the CAM and SE particles respectively, must be optimized. Ensuring that $\bar{D}_{\text{SE}}$ remains smaller than $\bar{D}_{\text{CAM}}$ is crucial for sustaining a percolating SE network at high f_{CAM} [14], which is the desired range for practical applications.

While many of these optimization effects can be assessed on a larger scale through measurements such as cycling performance, effective conductivities, active material utilization, and porosity, analyzing their impact on the microstructure at a finer, microscopic scale remains challenging. Advanced imaging techniques, such as X-ray computed tomography (CT) [16–18] and Focused Ion-Beam Scanning Electron Microscopy (FIB-SEM) [19,20], have provided valuable insights into these microstructures. However, these methods are often costly, time-consuming, and can lack the necessary resolution to fully understand the microstructural complexities. These techniques are typically employed to visualize and quantify porosity, tortuosity, particle distribution, and phase interfaces in composite cathodes. CT allows for non-destructive 3D imaging of the bulk microstructure, including cracks, pores, and particle connectivity, whereas FIB-SEM enables high-resolution cross-sectional analysis, particularly suited for assessing interfacial morphology and nanoscale contact quality. Despite their limitations, these imaging approaches offer valuable experimental data that can serve as inputs or validation targets for computational models such as DEM, bridging the gap between simulated and real microstructures.

To complement experimental analyses, microstructural modeling has become an essential tool for gaining deeper insights into the interactions within ASSB cathodes. Recent advances in modeling and simulation techniques, such as Finite-Element Method (FEM) and Discrete Element Method (DEM), have enabled detailed studies of cathode microstructures. These models have demonstrated that the microstructural properties of the cathode are heavily influenced by factors such as material choice, composition, particle sizes, and processing conditions [14, 15,21–24]. DEM, in particular, has proven to be a promising method for

studying the effects of mechanical properties on the cathode's morphology [25]. It allows for the simulation of individual particle interactions and provides insights into particle packing and rearrangement behavior [26]. Recent studies employing DEM have modeled percolation networks to understand the impact of particle size and distribution on ionic conduction [14], investigated calendaring processes with explicit consideration of CAM and carbon-binder domains [27], calibrated electrical conductivity models against experimental electrode data [28], examined the effects of cold-press fabrication conditions on mechanical degradation during intercalation [29], analyzed particle compaction and plastic deformation during cold pressing to establish correlations with ionic conductivity [30], and assessed how solid electrolyte coating structures influence transport properties and particle damage under compression [31].

Despite the proven potentials, DEM simulations can be highly sensitive to the variation of input parameters. Therefore, reliable measurements of material properties, such as particle size distribution, Young's modulus, and interparticle behavior, are essential for accurate modeling. For instance, particle size distribution can be measured using methods like X-ray diffraction particle size analyzers [32], while the elastic properties of materials can be assessed using techniques such as nanoindentation [33] or the ultrasonic pulse-echo method [34]. However, there is significant variability in the reported mechanical properties of these materials in the literature, largely due to differences in synthesis routes, measurement conditions and techniques. Additionally, in DEM microstructural models for ASSBs, parameters such as friction coefficients between particles are often chosen arbitrarily or adjusted to fit experimental data, which can affect the accuracy of the simulations [14,28–31]. Therefore, conducting a parameter sensitivity analysis to evaluate how these parameters influence particle rearrangement, packing, percolation behavior, and contact relations within multi-component and multi-particle sized systems is essential for accurate generation of DEM microstructures and further optimization of ASSB cathodes.

This study aims to systematically explore and address the critical challenges associated with optimizing the microstructure of ASSB cathodes by employing a comprehensive DEM simulation approach, combined with experimental validation. The primary focus of this work is to investigate the influence of various input parameters, such as particle size distribution, mechanical properties, and processing conditions, network formation between CAM and SE, porosity, and ionic percolation within composite cathodes. By conducting a detailed parameter sensitivity analysis, we seek to understand how these variables interact and affect the overall microstructure generation of the cathode. Additionally, this study provides an experimental comparison of simulated microstructures with real-world cathode samples, to validate the accuracy of the DEM model. The insights gained from this research are expected to contribute to the optimization of parameter settings in DEM models, enabling the accurate generation of microstructures and the prediction of critical performance indicators, such as porosity and ionic percolation, in ASSB composite cathodes.

The remainder is structured as follows: Section 2 explains the experimental methods employed in this research, focusing on the preparation and testing of the materials. Section 3 details the computational models and simulations. Section 4 presents the results of the experiments and simulations, followed by a discussion of their implications in Section 5. Finally, Section 6 concludes the paper with a summary of key findings and suggestions for future work.

2. Experimental

All material-related experiments were conducted in an inert argon (Ar) atmosphere inside a glovebox ($p(\text{H}_2\text{O})$ and $p(\text{O}_2) < 1$ ppm) at room temperature. Materials were stored in dry argon-filled containers during any handling outside the glovebox to preserve their integrity unless otherwise specified. Additionally, unless specified, all received materials were used as provided, without any pretreatment.

2.1. Materials

In this work, an argyrodite sulfide-based Solid Electrolyte (SE) $\text{Li}_{5.5}\text{PS}_{4.5}\text{Cl}_{1.5}$ (CIS, Korea) and uncoated pristine poly-crystalline Cathode Active Material (CAM) $\text{LiNi}_{0.5}\text{Mn}_{0.3}\text{Co}_{0.2}\text{O}_2$ (Nippon Chemical Industrial Co., LTD., Japan) were used as primary materials for conducting experiments.

2.2. Wet-ball-milling

To reduce the particle size of the SE, wet ball milling was conducted using a planetary ball mill (Pulverisette 7, Fritsch GmbH, Germany). For the milling procedure, a 45 ml zirconium oxide (ZrO_2) milling jar and 1 mm diameter ZrO_2 milling balls were employed. The milling was conducted with an SE-to-ball mass ratio of 7:27 g at 500 rpm for a total duration of 160 min. This consisted of alternating 10 min milling and 10 min rest intervals to allow cooling. Butyl acetate was used as the solvent, with a quantity of 20 ml. After the milling, the SE and solvent mixture were separated from the milling balls and dried in a Büchi oven B-585 (Büchi Labortechnik GmbH, Germany) at 100 °C for 24 h under vacuum. Following the drying, the milled SE was hand-ground using a mortar to break up any resulting agglomerates.

2.3. Particle size analysis

The particle size distribution (PSD) of the SE and CAM materials was measured using a laser diffraction particle size analyzer LA-960V2 (Horiba Ltd., Japan). To prepare the samples for analysis, a small quantity of each material was dispersed in butyl acetate, followed by ultrasonication for 5 min to ensure a uniform dispersion. The measurements were conducted outside of the glovebox. Given the brief measurement duration and dispersion in butyl acetate, it is assumed that there were no significant changes in particle size or morphology during the process.

2.4. Electrochemical impedance spectroscopy

To measure the ionic bulk conductivity of each SE, electrochemical impedance spectroscopy (EIS) was performed using a VMP-300 (Biologic, France). First, 200 mg of the SE was uniformly distributed inside a 10 mm inner diameter polyetheretherketone (PEEK) cylinder between two stainless steel rods, forming an ion-blocking symmetric cell. Afterwards, the SE was densified uniaxially at 350 MPa for 2 min. The assembled cell was then placed inside an appropriate airtight housing, and during measurement, a torque of 5.0 N m (approximately 40 MPa) was applied as stack pressure. The EIS measurements were conducted with a voltage amplitude of 10 mV across a frequency range of 1 Hz to 7 MHz.

2.5. Powder X-ray diffraction (XRD)

The X-ray diffraction patterns of the SE samples were measured with an X-ray powder diffractometer (Rigaku, Japan) with a CuK_α radiation. The sample was prepared inside the glovebox using an airtight sample holder. The diffraction data was collected over an angle range of $2\theta = 10^\circ$ – 80° with a step width of 0.01° .

2.6. Scanning electron microscopy (SEM)

The CAM and SE samples were analyzed using a Phenom Pure Desktop Scanning Electron Microscope (SEM). The sample preparation was conducted in a glovebox under an Ar-atmosphere to minimize exposure to air. The SEM analysis was performed at an accelerating voltage of 3 kV.

Table 1

Weight and volume ratios of cathode compositions with varying CAM:SE content and fixed additive content (CB, CNF, PTFE).

f_{CAM} (wt.%)	CAM:SE:CB:CNF:PTFE	
	wt.%	vol.%
60	60:37:1:1:1	34.3:61.6:1.4:1.4:1.3
70	70:27:1:1:1	45.0:50.5:1.5:1.6:1.4
80	80:17:1:1:1	58.6:36.2:1.8:1.8:1.6
85	85:12:1:1:1	66.9:27.4:1.9:2.0:1.8

2.7. Cathode dry-film (DF) preparation

A solvent-free, DF method was utilized to prepare free-standing cathode sheets. Beginning with a sequential mixing approach using a planetary ball mill (Pulverisette 7, Fritsch GmbH, Germany) equipped with a 45 ml airtight ZrO_2 jar and 5 mm ZrO_2 balls, a powder-to-ball ratio was maintained at 1:5. Initially, the CAM and SE were milled for 15 min in total (5 min of mixing followed by 5 min of rest) at 300 rpm. Then, CA comprising of CB Super C45 (Imerys, France) and CNF were added, and the mixture was milled for 10 min in total at 200 rpm (5 min cycles of mixing and resting). Finally, the PTFE binder (Chemours, United States) was added, and the mixture was milled for an additional 5 min. The resulting mixture was transferred to a preheated (100 °C) mortar grinder (Pulverisette 2, Fritsch GmbH, Germany) and processed (approximately 2 min), until a dough-like flake was formed. The resulting flake was then rolled out on a 100 °C preheated template to achieve the desired thickness (approximately 105 μm), resulting in a freestanding cathode sheet. Different cathode compositions were prepared, varying the CAM:SE ratios (60:37, 70:27, 80:17, and 85:12 by weight) while keeping equal proportions of each CB, CNF, and PTFE constant at 1 wt% each. The complete compositions, including both weight and corresponding volume ratios, are summarized in Table 1. Each composition was prepared using two different SE particle sizes while keeping the CAM particle size distribution constant across all samples.

2.8. Cell fabrication and testing

For electrochemical testing, torque cells were fabricated composed of a PEEK cylinder with an inner diameter of 10 mm and two 10 mm diameter stainless steel plunger as current collectors. Initially, 160 mg of pristine SE was evenly distributed inside the PEEK cylinder and uniaxially pressed at 100 MPa for 1 min to form a separator layer. A 9 mm coin, punched from a freestanding cathode sheet, was then centered on top of the SE layer, and further densified at either 200 or 400 MPa for 1 min. Then, a Lithium–indium alloy anode, consisting of a 10 mm indium coin with a thickness of 100 μm (Carl Roth GmbH + Co. KG, Germany) and a 9 mm lithium coin with a thickness of 30 μm , was attached to the opposite side of the SE layer, with the indium side facing the SE separator. Following this, the assembled cell was then sealed in an Ar-filled airtight housing and subjected to cycling under a constant torque of 5.0 N m (approximately 40 MPa) at 25 °C. All cycling tests were conducted using a VMP-300 (Biologic, France) within a voltage window of 3.6–1.9 V versus Li-In for charge and discharge respectively. The cell was charged using a constant current constant voltage (CCCV) protocol with a 5-h duration of the CV phase at an upper cutoff voltage of 3.6 V, and discharged with a constant current (CC) step. The current density was set at 0.05 mA cm^{-2} . This relatively low current density was intentionally chosen to minimize kinetic limitations during electrochemical cycling, enabling the CAM utilization (θ_{CAM}) to be evaluated under conditions where rate-dependent losses are negligible.

Table 2
Material densities used in this work.

Description	Parameter	Unit	Value	Source
Density (CAM)	ρ_{CAM}	g cm^{-3}	4.77	[35]
Density (SE)	ρ_{SE}	g cm^{-3}	1.64	[36]
Density (CB)	ρ_{CB}	g cm^{-3}	2.00	[37]
Density (CNF)	ρ_{CNF}	g cm^{-3}	1.90	Supplier
Density (PTFE)	ρ_{PTFE}	g cm^{-3}	2.15	[38]

2.9. Experimental θ_{CAM} determination

The CAM utilization (θ_{CAM}) was calculated based on electrochemical discharge measurements. The method for determining utilization followed the approach reported in [14], where the highest experimentally measured specific discharge capacity is used as the reference point corresponding to 100% utilization. In this study, a value of 150 mAh g^{-1} , obtained for the SE19-based composite with 60 wt.% CAM at 400 MPa (see Fig. 15), was selected as the reference capacity. All other measured discharge capacities were normalized against this value to calculate θ_{CAM} .

The experimental utilization was calculated using the following relation:

$$\theta_{\text{CAM}} = \frac{C_{\text{spec. discharge}}}{C_{\text{max. discharge}}} \quad (1)$$

where $C_{\text{spec. discharge}}$ is the measured specific discharge capacity, and $C_{\text{max. discharge}} = 150 \text{ mAh g}^{-1}$. As this approach relies on a single reference capacity as the benchmark for 100% utilization, deviations in the reference measurement or variability in experimental reproducibility could influence the normalization. However, using a clear benchmark enables consistent comparisons across the tested conditions.

2.10. Geometrical porosity determination

The geometrical porosity of the pressed cathodes was determined by first preparing DF cathodes, which were uniaxially pressed at either 200 or 400 MPa for 1 min. The pressed cathodes were then punched into 9 mm diameter coins. These coins were subsequently weighed, and their thickness was measured to calculate the density (ρ) of the cathode using:

$$\rho_{\text{measured}} = \frac{m}{A \cdot h} = \frac{m}{\pi \cdot (d/2)^2 \cdot h} \quad (2)$$

where m is the mass of the cathode, d is the diameter (9 mm), $A = \pi \cdot (d/2)^2$ is the area, and h is the measured thickness.

The porosity (ϵ) of the cathodes was then determined using the formula:

$$\epsilon = \left(1 - \frac{\rho_{\text{measured}}}{\rho_{\text{theoretical}}} \right) \quad (3)$$

where ρ_{measured} is the measured density of the cathode after pressing and $\rho_{\text{theoretical}}$ is the theoretical density of the cathode. For clarity, the term “porosity” in this context refers specifically to the bulk voids within the cathode and does not account for interparticle voids, such as those found in polycrystalline materials. All densities of the materials used are listed in Table 2. It should be noted that this method depends on manual measurements of thickness and mass, which may introduce variability and lead to deviations in absolute porosity values.

3. Cathode composite microstructural modeling

3.1. DEM model equations

To generate the cathode composite microstructures for the investigations in this study, we use the DEM model which is a physics-based approach considering the interactions between the CAM and SE

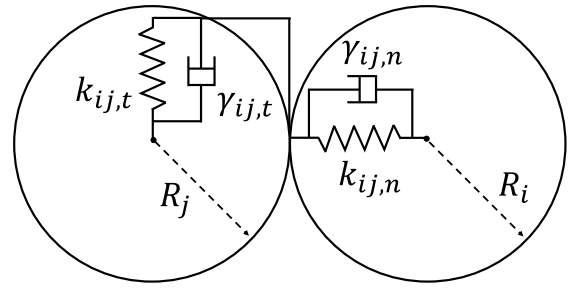


Fig. 1. Schematics of the used spring–dashpot contact model, defined according to Eqs. (6) and (7).

particles. This approach enables generating different stochastic representation of the cathode microstructure, given the CAM and SE particle size distributions, their mechanical properties, and the applied pressure during the cold pressing process. In the DEM as a Lagrangian method, the computational domain is treated as assemblies of discrete particles, where their trajectory is explicitly defined based on the Newton's laws of motion. Complex particle shapes can also be used in the model by forming agglomerates of multiple spherical particles or other well-defined geometrical shapes [25,31,39]; however, to minimize the associated computational costs, in this work the SE and NMC-cathode particles are considered as spheres.

The physical governing equations for a particle with index i , with position $\mathbf{x}_i$, mass m_i , rotational velocity ω_i and moment of inertia I_i , are the force balance equations considering contact force vectors from neighbor particles with index j ($\mathbf{F}_{ij}$) and other external forces ($\mathbf{F}_i^{\text{ext}}$) such as gravitational force which read

$$m_i \ddot{\mathbf{x}}_i = \sum_j \mathbf{F}_{ij} + \mathbf{F}_i^{\text{ext}} \quad (4)$$

$$I_i \dot{\omega}_i = \sum_j \mathbf{r}_i \times \mathbf{F}_{ij} + \mathbf{T}_i \quad (5)$$

where $\mathbf{r}_i$ represents the particle's radius, and $\mathbf{T}_i$ is an additional torque which can be used to model the particles' non-sphericity using a rolling friction method [40].

The contact force between two colliding particles in this work is calculated based on a spring–dashpot model [14,40,41]. The normal and tangential components of the contact force, denoted as $\mathbf{F}_{ij,n}$ and $\mathbf{F}_{ij,t}$, are respectively calculated using Eqs. (6) and (7), schematically illustrated in Fig. 1.

$$\mathbf{F}_{ij,n} = k_{ij,n} \delta_{ij,n} - \gamma_{ij,n} \mathbf{u}_{ij,n} \quad (6)$$

$$\mathbf{F}_{ij,t} = k_{ij,t} \delta_{ij,t} - \gamma_{ij,t} \mathbf{u}_{ij,t} \quad (7)$$

The elastic term in the normal direction is modeled by a Hertzian spring, and the damping term is represented by an additional spring in the tangential direction. These terms in the employed contact model are functions of the normal and tangential components of the displacement δ_{ij} and relative velocity $\mathbf{u}_{ij}$ vectors, where $k_{ij,n}$ and $k_{ij,t}$ are respectively the normal and tangential spring coefficients, and $\gamma_{ij,n}$ and $\gamma_{ij,t}$ are the viscoelastic damping factors which are functions of particles' overlap and material properties, the details of which are described in Appendix. It should be noted that the tangential elastic force calculated in Eq. (7) is truncated to fulfill the Coulomb friction criterion [40], where the magnitude of the tangential force cannot exceed μ_{ij} times the normal force magnitude. See Appendix for the detailed mathematical description of the truncation procedure.

A similar contact model is also applicable for the particle–wall contacts [40,41], as it is described in Appendix. At each time-step, the model searches and updates all particle–particle and particle–wall contacts, calculates corresponding contact forces, and solves the dynamic governing equations per particle through a finite-difference scheme.

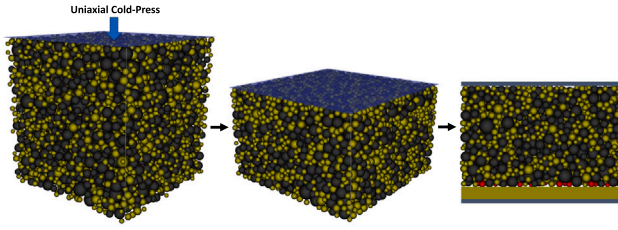


Fig. 2. Snapshot of the cathode composite microstructure generation process. The CAM (black) and SE (yellow) particles are initially randomly distributed in a cubic box where they undergo a uniaxial cold press. When the granular system fully settles down after reaching the prespecified pressure, the achieved microstructural data is used to perform the network-based analysis explained in Section 3.3, considering the target SE particles (red) that are in contact with the SE separator layer (yellow sheet).

3.2. Microstructural modeling setup

In this work, the DEM model is implemented in the open-source package LIGGGHTS [40,42]. The computational domain of the granular system is defined as a cubic box with periodic boundary conditions at the side walls, a fixed boundary at the bottom and a downward-moving servo wall at the top which resembles a uniaxial cold press. At the initial state, the CAM and SE particles are randomly inserted into the simulation box, overlap-free, with respect to their corresponding particle size distribution represented by multiple bins per particle type. The mass fractions of the CAM and SE particles in the model are defined based on the cathode loading and the mass fraction of CB (f_{CB}), CNF (f_{CNF}) and PTFE binder (f_{PTFE}) phases. As the primary objective of this study is to investigate the ionic transport in the cathode composite, for minimizing the computational costs, the additives are not explicitly included in the model, nevertheless the effect of total additives mass fraction (f_{add}) on the overall mass of the system (m_{tot}) is considered as

$$m_{tot} = \frac{m_{CAM} + m_{SE}}{1 - (f_{CB} + f_{CNF} + f_{PTFE})} = \frac{m_{CAM} + m_{SE}}{1 - f_{add}} \quad (8)$$

Therefore, the adjusted mass fractions of CAM and SE particles in the DEM model equal $\frac{f_{CAM}}{1-f_{add}}$ and $1 - \frac{f_{CAM}}{1-f_{add}}$, respectively.

The particles in the model are subjected to gravitational force, and the upper wall is moving downward with a maximum velocity of $0.2 \mu\text{m}/\mu\text{s}$, controlled by a Proportional–Integral–Derivative (PID) controller to achieve a predefined total target pressure on the moving surface. Then, the target uniaxial pressure is held until the total kinetic energy of the system tends to zero; practically falling below a specified threshold. To assure the solution stability, the time-steps are set so that to fulfill the Rayleigh and Hertz time-step criteria for all particles in the system [43].

The generated composite cathode microstructures, are used as an input to a network-based post processing tool developed as a MATLAB code to calculate the objective CAM- and SE-Utilization factors, the details of which are explained in Section 3.3. The input scripts for the DEM simulations in LIGGGHTS are also generated using a MATLAB code, which enables automating the whole workflow to generate large amount of data, as well as providing the potential to perform a model-based optimization study. The microstructure generation process in this work is illustrated in Fig. 2, depicting the studied granular system at the initial state, final state, and resulting electrode used for further investigations.

To account for statistical variations of the randomly generated microstructural data, in this work three different simulations are performed per case study. In the rest of this paper the mean value of those realizations are used for comparison purposes, and the corresponding error bars in the figures are based on the calculated standard deviation per case study. In Section 4.2, for different cathode loadings, we

have performed a detailed convergence study in terms of the observed variations in the computed electrodes' porosity and utilization factors while changing the computational domain size. This is followed by providing practical suggestions to define a proper Representative Volume Element (RVE) to obtain meaningful results. The DEM simulations in this study are executed as multiple tasks on a Lenovo ThinkSystem SR650, operating under SUSE Linux Enterprise Server.

3.3. CAM- and SE-utilization

To quantify the CAM-utilization (θ_{CAM}), we followed a methodology similar to Shi et al. [14]. The output θ_{CAM} was determined by analyzing Li-ion percolation pathways within the SE matrix, where Li-ion diffusion is significantly faster than in the CAM [44,45]. To identify active CAM particles, we constructed a network model, treating each CAM and SE particle as a node connected to neighboring particles. Two particles were considered neighbors if the distance between their centers was less than the sum of their radii. CAM particles were designated as active if connected through at least one percolation pathway to the bulk SE layer at the bottom of the microstructure [46]. θ_{CAM} was computed as the ratio of the active CAM particle volume (V_{CAM}^{active}) to the total CAM volume (V_{CAM}^{total}) in the system:

$$\theta_{CAM} = \frac{V_{CAM}^{active}}{V_{CAM}^{total}} \quad (9)$$

For SE-utilization (θ_{SE}), a similar approach was employed by isolating the SE network from the CAM particles to focus solely on SE connectivity. θ_{SE} was calculated by evaluating the SE network's connectivity to the bulk SE layer. This involved identifying SE particles linked in continuous pathways leading to the separator and calculating the fraction of SE volume in these pathways versus those that were isolated. The equation for θ_{SE} is:

$$\theta_{SE} = \frac{V_{SE}^{connected}}{V_{SE}^{total}} \quad (10)$$

where $V_{SE}^{connected}$ is the volume of SE particles that are connected in continuous pathways leading to the separator, and V_{SE}^{total} is the total SE particle volume in the system.

3.4. Porosity calculation with Monte Carlo approach

In our DEM simulations, the use of soft spheres results in particle overlaps, complicating the direct calculation of porosity. Analytical methods to account for these overlaps would be inefficient and time-consuming. Therefore, we utilized a Monte Carlo approach, as described in [47], to analyze the packing behavior of the generated microstructures and calculate porosity. Random Monte Carlo points were inserted into the simulation domain to determine if each point fell within a CAM particle, SE particle, or a void. The volume fractions of each component were then calculated based on the proportion of points in each category:

$$v_{CAM} = \frac{N_{CAM}}{N_{total}} \quad (11)$$

$$v_{SE} = \frac{N_{SE}}{N_{total}} \quad (12)$$

$$v_{void} = \frac{N_{void}}{N_{total}} \quad (13)$$

where v_{CAM} , v_{SE} , and v_{void} represent the volume fractions of the CAM, the SE, and the voids respectively, and N_{CAM} , N_{SE} , N_{void} are the numbers of points that fall inside CAM, SE, and void regions respectively, and N_{total} is the total number of Monte Carlo points. Given the significantly higher Young's modulus of CAM compared to SE (Table 2), any points within the overlap between SE and CAM particles were

Table 3
DEM simulation parameters used in this work.

Description	Parameter	Unit	Boundaries	Nominal value	Source
SE-SE friction coefficient	μ_{SE-SE}	–	0.2–0.6	0.4	This work
CAM-CAM friction coefficient	$\mu_{CAM-CAM}$	–	0.2–0.6	0.4	This work
CAM-SE friction coefficient	μ_{CAM-SE}	–	0.2–0.6	0.4	This work
SE Young's modulus	E_{SE}	GPa	15–26	20.5	[14,31,34,48,49]
CAM Young's modulus	E_{CAM}	GPa	123–200	161.5	[14,31,33–35,50–54]
SE Poisson ratio	ν_{SE}	–	0.29–0.37	0.33	[14,31,48,55]
CAM Poisson ratio	ν_{CAM}	–	0.24–0.3	0.27	[14,31,33,35,51]
Coefficient of restitution	RC	–	0.2–0.6	0.4	This work
Pressure	p	MPa	200–400	300	This work

considered as CAM. The volumes of CAM (V_{CAM}), SE (V_{SE}) and voids (V_{void}) were calculated using their respective volume fractions:

$$V_{CAM} = v_{CAM} \cdot V_{box} \quad (14)$$

$$V_{SE} = v_{SE} \cdot V_{box} \quad (15)$$

$$V_{void} = v_{void} \cdot V_{box} \quad (16)$$

where V_{box} is the volume of the analyzed simulation domain. The mass of CAM (m_{CAM}) and SE (m_{SE}) was then calculated using their respective densities (ρ_{CAM} and ρ_{SE}):

$$m_{CAM} = V_{CAM} \cdot \rho_{CAM} \quad (17)$$

$$m_{SE} = V_{SE} \cdot \rho_{SE} \quad (18)$$

With m_{total} known (based on Eq. (8)), the volume contributions of each additive (V_{CB} , V_{CNF} , and V_{PTFE}) were calculated using their known densities ρ_{CB} , ρ_{CNF} , and ρ_{PTFE} and their mass fractions $f_{CB} = 0.01$, $f_{CNF} = 0.01$, and $f_{PTFE} = 0.01$:

$$V_{CB} = \frac{f_{CB} \cdot m_{total}}{\rho_{CB}} \quad (19)$$

$$V_{CNF} = \frac{f_{CNF} \cdot m_{total}}{\rho_{CNF}} \quad (20)$$

$$V_{PTFE} = \frac{f_{PTFE} \cdot m_{total}}{\rho_{PTFE}} \quad (21)$$

Finally, the correct porosity (ϵ) is calculated by subtracting the volume contributions of the additives from the initial void volume:

$$\epsilon = \frac{V_{void} - (V_{CB} + V_{CNF} + V_{PTFE})}{V_{box}} \quad (22)$$

3.5. Parameter set for DEM model

The simulation parameters employed in this study are detailed in Table 3. Most of these parameters were sourced from literature and previous research efforts.

3.6. Parameter sensitivity analysis

A parameter sensitivity analysis was conducted on all eight parameters (μ_{SE-SE} , $\mu_{CAM-CAM}$, μ_{CAM-SE} , E_{SE} , E_{CAM} , ν_{SE} , ν_{CAM} and RC) to evaluate their impact on model outputs θ_{CAM} , θ_{SE} , and ϵ . Using the parameter ranges established in Table 3, we assessed each parameter's sensitivity by examining the changes in model outputs as individual parameters were varied. The analysis was performed under different pressures (200, 300, and 400 MPa), across different f_{CAM} (60, 70, 80, and 85 wt% CAM) and two different SE particle size distributions, while keeping the CAM particle size distribution constant across all simulations.

This comprehensive sensitivity analysis allowed us to identify the most influential parameters affecting θ_{CAM} , θ_{SE} , and ϵ under varying mechanical and compositional conditions. To perform this sensitivity analysis, we employed a One-At-a-Time (OAT) method, as described

in [56]. For each parameter, 10 value points were selected under a discrete uniform distribution between the lower and upper boundaries, while the remaining parameters were held constant at their nominal values. The i th value point q_i for each parameter q is calculated as:

$$q_i = q_l + (i - 1) \cdot \frac{(q_u - q_l)}{N_s - 1} \quad (23)$$

where q_l and q_u represent the lower and upper boundaries of the parameter q , and $N_s = 10$ is the number of value points. The nominal value $\bar{q}$ of each parameter q is given by:

$$\bar{q} = \frac{q_l + q_u}{2} \quad (24)$$

To quantify the sensitivity, we calculated the sensitivity index (SI) for each parameter, defined as the standard deviation of the analyzed state x_i at specific points [57]:

$$SI = \sqrt{\frac{1}{N_s} \sum_{j=1}^{N_s} (x_{ij} - \bar{x}_i)^2} \quad (25)$$

where x_{ij} represents the state corresponding to the j th value point of the i th parameter, and $\bar{x}_i$ is the average value of the states x_{ij} .

3.7. Model assumptions and simplifications

In the DEM model used, several assumptions were made to ensure computational efficiency and feasibility. As discussed in Section 3.1, all particles were assumed to be spherical to simplify interactions. Particles were randomly inserted to achieve a homogeneous distribution, which may overlook potential clustering effects resulting from mixing, potentially leading to an overestimation of connectivity, and therefore θ_{CAM} and θ_{SE} . Interparticle interactions were modeled using idealized spring-dashpot contact models, without accounting for complexities such as plastic deformation and adhesive behavior, possibly leading to an underestimation of contact areas between particles. The presence of additives (CB, CNF, and PTFE) was not directly simulated but approximated using a pre-factor. As a result of these assumptions, the model may not perfectly capture the actual electrode behavior and may deviate from exact experimental values. Therefore, the model's results should be interpreted cautiously, acknowledging the inherent limitations introduced by these necessary simplifications.

4. Results

4.1. Material properties

The SEM images of the CAM, pristine SE (SE45), and size-reduced SE (SE19) after wet ball milling are shown in Fig. 3(a–c), respectively, providing a visual representation of the particle morphologies. The CAM exhibits a relatively smooth morphology, while SE45 and SE19 show different particle size distributions due to the reduction in size of SE19 via wet ball milling.

The particle size distribution (PSD) of the CAM, pristine SE (SE45), and the size-reduced SE (SE19) are shown in Fig. 4(a). The CAM exhibits a mean diameter ($\bar{D}_{CAM}$) of 5.8 μm , while SE45 shows a

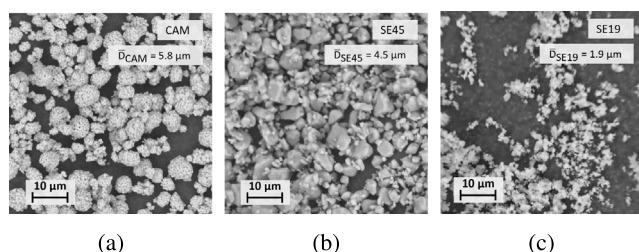


Fig. 3. SEM images of (a) CAM, (b) pristine SE (SE45), and (c) wet ball milled SE (SE19).

mean diameter ($\bar{D}_{SE45}$) of 4.5 μm . After wet ball milling, the SE particle size was significantly reduced, resulting in a mean diameter ($\bar{D}_{SE19}$) of 1.9 μm , and this size-reduced SE will be referred to as SE19. While the SEM image of the pristine SE Fig. 3(b) shows a visible presence of smaller particles alongside larger ones, which may give an impression of a bimodal distribution, the laser diffraction measurement reflects a predominantly unimodal, Gaussian-like particle size distribution. This apparent discrepancy arises from the differences in measurement principles: SEM provides a qualitative surface view, whereas laser diffraction measures the volume-weighted distribution of the full sample. Notably, although particles smaller than 1 μm were observed in SEM and captured in the laser diffraction data, their contribution to the overall volume fraction is very small (<2%), explaining their minimal influence on the bulk PSD curve. Fig. 4(b) displays the EIS spectra for both SE45 and SE19. The extracted bulk ionic conductivity for SE45 was 2.9 mS cm^{-1} , whereas SE19 exhibited a reduced conductivity of 1.0 mS cm^{-1} . This decrease in ionic conductivity for SE19 is consistent with previous observations in similar types of solid electrolytes, where extended milling times introduce additional grain boundaries, impeding ionic transport and reducing overall conductivity [14,58]. Fig. 4(c) presents the XRD patterns for both SE19 and SE45. Both samples exhibit crystalline behavior typical of this material class, with sharp and well-defined peaks indicating a high degree of crystallinity [59]. The similarity in the XRD patterns between SE19 and SE45 suggests that the crystalline structure of the material remained largely intact despite the reduction in particle size through wet ball milling. The absence of additional peaks or significant shifts further indicates that no new phases or structural irregularities were introduced during the milling process.

4.2. Simulation box size and representative volume element (RVE) determination

To assess the influence of simulation box size (Box Size Factor) on θ_{CAM} , θ_{SE} and ϵ , we analyzed the various f_{CAM} (60, 70, 80, and 85 wt% CAM) using SE19 and SE45. The simulation employed the nominal parameter values listed in Table 3, with a pressure of 300 MPa, representing the nominal pressure within the range being analyzed.

As shown in Fig. 5, both θ_{CAM} and θ_{SE} for SE19 (Fig. 5(a) and (b)) and SE45 (Fig. 5(c) and (d)) converge and stabilize from a box size factor of 10, consistent with findings reported in [14]. Fig. 6 demonstrates that ϵ steadily decreases with increasing box size for both SE19 (Fig. 6(a)) and SE45 (Fig. 6(b)) across all f_{CAM} . This decrease can be attributed to the “wall effects” typical in DEM microstructure simulations (Fig. 7(a)), where particles near the walls arrange more densely, leading to higher packing densities near the walls and a more random arrangement farther away [60]. In our model, periodic boundary conditions are applied in the horizontal directions, effectively eliminating side-wall effects. However, the bottom wall (fixed boundary) and top surface (free surface under compression) remain physical boundaries and are the primary sources of observed wall effects in this study. Additionally, the presence of an uneven free surface, particularly at

the simulation box boundaries, contributes to fluctuations in the void ratio, further affecting overall porosity. The observed decrease in ϵ in Fig. 6 aligns with the understanding that particles within 2–4 diameters of the walls should not represent bulk consolidation, especially in spherical systems [61–63]. Thus, the diminishing influence of wall effects with larger box sizes leads to a more representative bulk porosity measurement.

To accurately determine the porosity ϵ , a virtual box with a side length ratio (ω) of 1 was defined. By progressively decreasing ω , regions near the top and bottom boundaries were excluded to reduce the influence of vertical wall effects (Fig. 7(b)). By varying ω from 0.1 to 1 for SE19 and SE45 at different f_{CAM} , we identified a plateau region ($\omega \approx 0.4 - 0.9$) where ϵ remains constant, indicating a sufficiently large RVE (Fig. 8(a), (b)), aligning with the convergence behavior observed in [60]. Beyond this plateau, minor increases in ϵ suggest residual top and bottom wall effects. Therefore, $\omega = 0.7$ was selected for all subsequent simulations, as it effectively balanced ϵ representation.

The influence of the box size factor on ϵ was re-evaluated with this adjusted side length ratio. The results (Fig. 9) confirmed that ϵ stabilizes beyond a box size factor of 8 with $\omega = 0.7$, indicating that the chosen parameters provide a reliable representation of the cathode's porosity, free from boundary effects.

To ensure consistent results, we varied the number of Monte Carlo points from 100 to 200,000 (approximately ~ 0.008 to ~ 16.2 points per μm^3). Fig. 10(a) and (b) for SE19 and SE45, respectively, show that after 30,000 points, the outcomes are consistent and stable. This suggests that using 30,000 Monte Carlo points is sufficient to achieve precise and dependable porosity calculations, ensuring that the calculated volume fractions of CAM, SE, and voids are accurate and representative of the microstructure.

Fig. 11 illustrates the simulation times as a function of the box size factor. The results demonstrate that SE19 simulations require significantly more computational time due to smaller particle sizes, leading to higher particle counts. Lower f_{CAM} further extends simulation times due to the dominance of SE particles. A box size factor of 10, combined with $\omega = 0.7$, was chosen for θ_{CAM} , θ_{SE} and ϵ calculations, balancing accuracy, computational efficiency, and feasibility. Despite the analysis indicating that 30,000 Monte Carlo points suffice for the porosity estimation, 50,000 points were used to ensure precise porosity calculations while managing computational demands.

4.3. Parameter sensitivity analysis

A total of 1920 simulation cases were conducted, varying two SE particle sizes (SE19 and SE45), four f_{CAM} (60, 70, 80, 85 wt%), three pressures (200, 300, 400 MPa), and eight parameters across ten discrete steps. Each case involved at least 3 simulation runs to ensure statistical reliability, with mean values and standard deviations calculated. Nominal values from Table 3 were applied to parameters not under investigation.

The sensitivity of the outputs θ_{CAM} , θ_{SE} , and ϵ was analyzed with respect to pressure and f_{CAM} . A strong correlation was found between f_{CAM} and the θ_{CAM} and θ_{SE} . At lower f_{CAM} , results showed minimal variation, suggesting full utilization of both CAM and SE due to a well-formed SE ionic percolation network. This is illustrated in Fig. 12(a–d), which show θ_{CAM} and θ_{SE} for SE45 at 400 MPa. In these examples, we specifically plotted the effects of the CAM-SE friction coefficient (μ_{CAM-SE}) and Young's modulus of SE (E_{SE}) to highlight the trends. Similar behavior was observed across other parameters (μ_{SE-SE} , $\mu_{CAM-CAM}$, E_{CAM} , ν_{SE} , ν_{CAM} , and RC) and pressures. These trends effectively represent the overall sensitivity of θ_{CAM} and θ_{SE} at lower f_{CAM} (<80 wt% CAM for SE45 and <85 wt% CAM for SE19), indicating sufficient percolation at these loadings.

However, as f_{CAM} increased and approached the percolation threshold, both θ_{CAM} and θ_{SE} decreased with rising μ_{CAM-SE} and E_{SE} values.

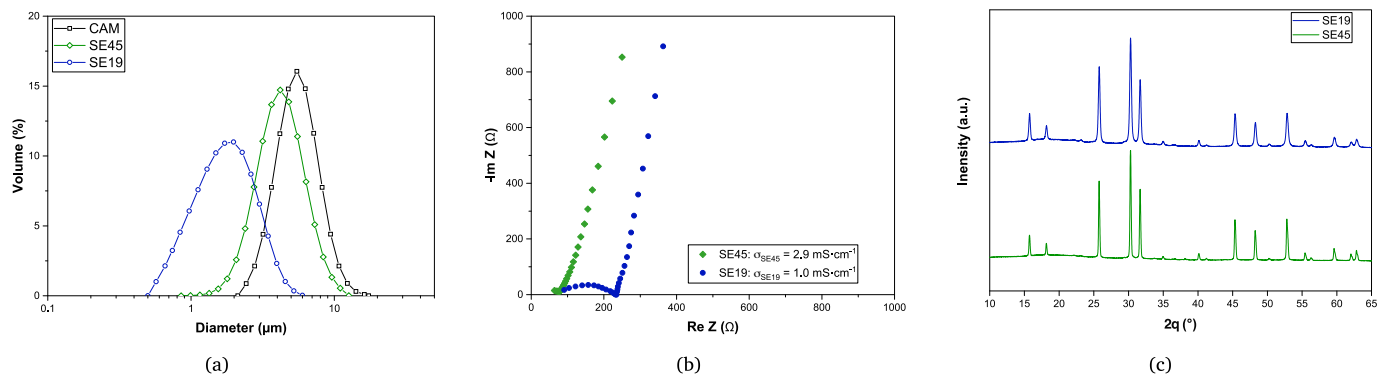


Fig. 4. (a) Particle size distribution of CAM, SE45, and SE19 materials, (b) EIS spectra of SE45 and SE19, and (c) XRD patterns of SE45 and SE19.

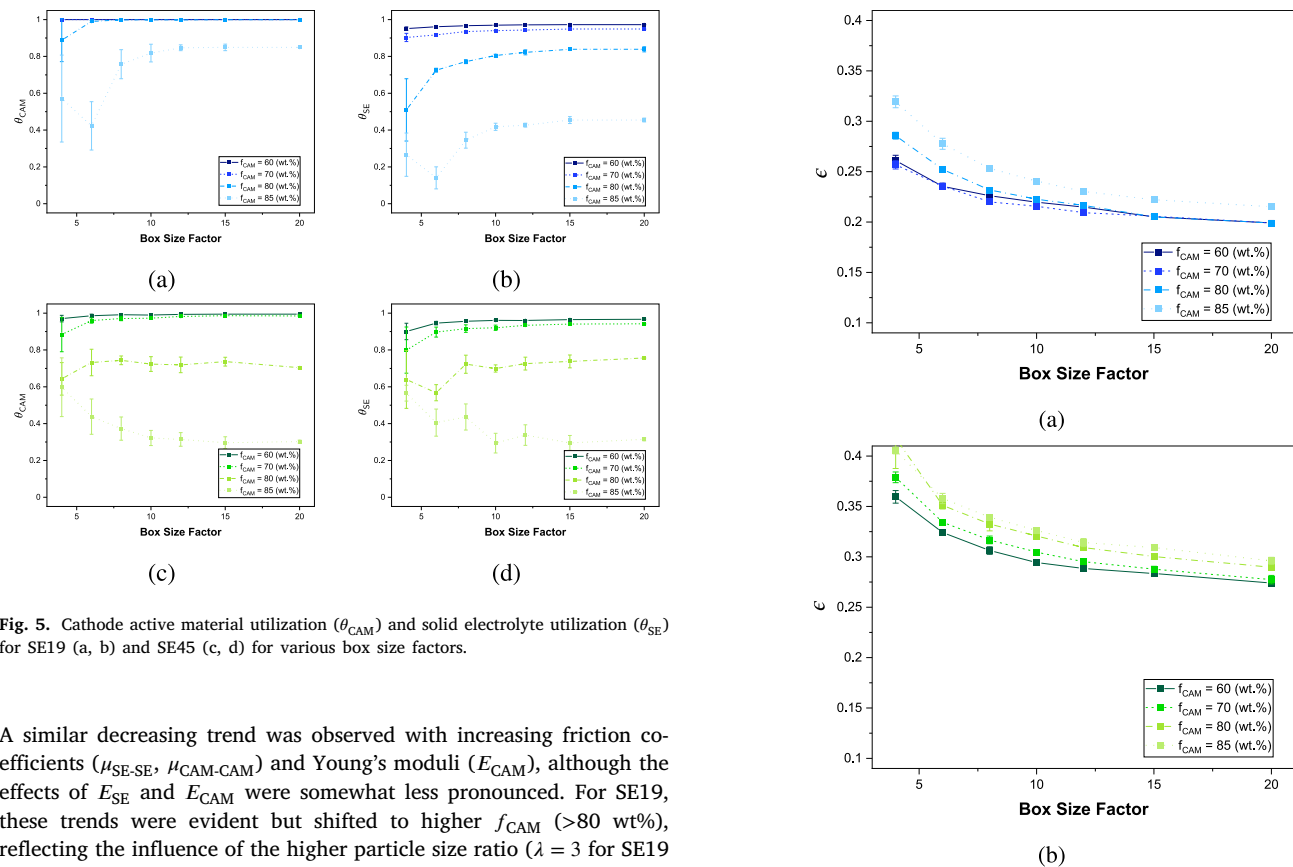


Fig. 5. Cathode active material utilization (θ_{CAM}) and solid electrolyte utilization (θ_{SE}) for SE19 (a, b) and SE45 (c, d) for various box size factors.

A similar decreasing trend was observed with increasing friction coefficients (μ_{SE-SE} , $\mu_{CAM-CAM}$) and Young's moduli (E_{CAM}), although the effects of E_{SE} and E_{CAM} were somewhat less pronounced. For SE19, these trends were evident but shifted to higher f_{CAM} (>80 wt%), reflecting the influence of the higher particle size ratio ($\lambda = 3$ for SE19 compared to $\lambda = 1.3$ for SE45) [14]. A higher λ facilitates improved percolation and utilization at higher f_{CAM} , thereby shifting the percolation threshold to a higher f_{CAM} , allowing for better connectivity and utilization of both the SE and CAM in the cathode.

Porosity (ϵ) exhibited different behavior, relating more to packing dynamics than percolation. As shown in Fig. 12(e) and (f), increases in μ_{CAM-SE} and E_{SE} led to higher porosity. This trend was consistent across all friction coefficients and Young's moduli, as well as for SE19, indicating a universal influence on packing density across different parameter settings.

Three parameters – namely the Poisson's ratios for SE and CAM (ν_{SE} , ν_{CAM}) and the Coefficient of Restitution (RC) – showed no significant trends. While some variations were observed, these are likely due to the random initial particle positions in each simulation run, rather than a direct impact of these parameters.

The sensitivity matrices in Fig. 13(a) and (b) were normalized between 0 and 1 for consistent comparison across parameters, CAM loadings, and pressures. These matrices reveal key trends: θ_{CAM} and

Fig. 6. Effect of box size factor on porosity (ϵ) at different cathode active material loadings (f_{CAM}) for (a) SE19 and (b) SE45.

θ_{SE} exhibited increased sensitivity at higher f_{CAM} , likely due to the diminishing effectiveness of the SE percolation network as CAM content increased. Lower f_{CAM} showed minimal sensitivity, indicating a well-connected network at these levels, where sufficient SE is available to form a percolating SE network connecting all CAM particles.

For ϵ , sensitivity varied with friction coefficients and Young's modulus. The friction coefficient between SE particles μ_{SE-SE} was particularly sensitive at lower f_{CAM} , reflecting the dominance of SE content. Conversely, $\mu_{CAM-CAM}$ was more sensitive at higher f_{CAM} , where CAM content dictated packing behavior. μ_{CAM-SE} displayed minimal sensitivity differentiation for SE45, but for SE19, sensitivity increased significantly at higher f_{CAM} , highlighting the impact of smaller, more numerous SE particles on CAM-SE interactions. RC showed no clear

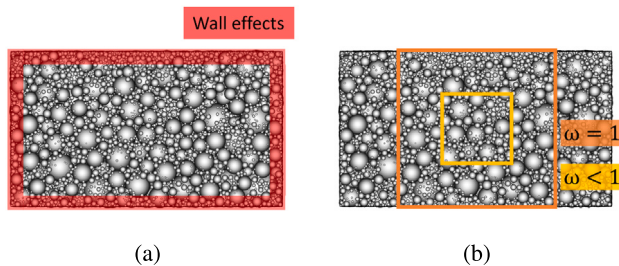


Fig. 7. (a) Illustration of wall effects, (b) definition of side length ratio (ω).

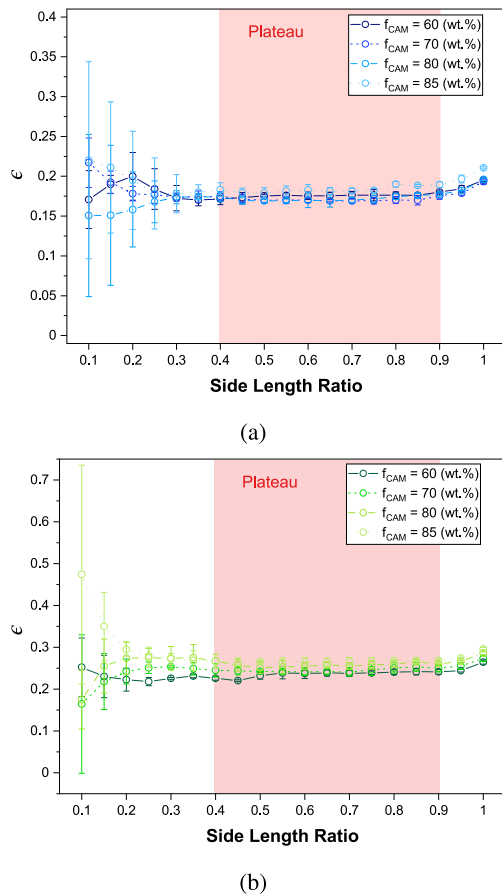


Fig. 8. (a) Porosity (ϵ) as a function of ω for SE19, (b) for SE45.

sensitivity trends, likely due to random variations in particle insertion during simulations. The Young's modulus E_{SE} was more sensitive at lower f_{CAM} , while E_{CAM} became more critical at higher f_{CAM} , corresponding to the increased influence of CAM content. Poisson's ratios for both SE (ν_{SE}) and CAM (ν_{CAM}) showed no significant trends, possibly due to their narrow range in literature, limiting their impact within this study. Despite pressure affecting packing and percolation behavior, the sensitivity matrices showed no significant pressure-related trends, suggesting that within the studied range, pressure variations have a limited impact compared to other factors like friction coefficients and Young's modulus.

Fig. 14(a) and (b) rank the normalized sensitivity indices for the eight parameters regarding their influence on θ_{CAM} , θ_{SE} , and ϵ for SE45 and SE19. The friction coefficient μ_{CAM-SE} consistently exhibited the highest sensitivity, underlining its critical role in utilization rates. Other friction coefficients also ranked highly, particularly for SE45, while Young's moduli displayed moderate sensitivity. Poisson ratios

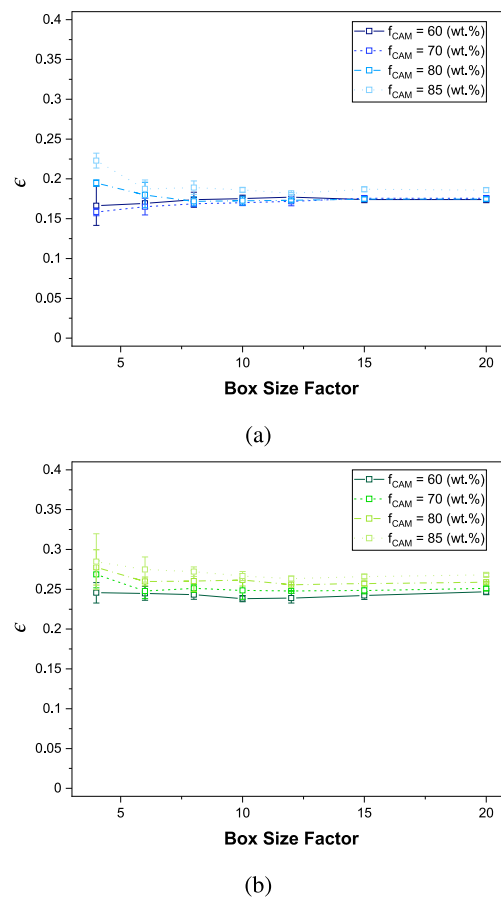


Fig. 9. Effect of box size factor on porosity (ϵ) with $\omega = 0.7$ for (a) SE19, (b) SE45.

and RC showed the lowest sensitivity, indicating minimal influence on percolation and utilization behaviors.

For ϵ , while μ_{CAM-SE} remained the most sensitive parameter, E_{SE} also showed significant sensitivity in SE45. For SE19, the importance of μ_{SE-SE} increased due to the smaller SE particles, while $\mu_{CAM-CAM}$ sensitivity decreased, reflecting the reduced influence of CAM-CAM interactions in a SE-dominated system. Poisson ratios and RC consistently had the lowest sensitivity across all cases.

Overall, SE45 showed greater sensitivity to parameter variations than SE19, likely due to the lower particle size ratio λ , making the system more responsive to changes in friction and mechanical properties. For porosity, larger SE particles in SE45 led to more pronounced packing effects, where parameter variations had a greater impact on overall porosity.

By systematically varying input parameters – such as particle size ratio, elastic properties, and friction coefficients – and quantifying their influence on key outputs like active material utilization and porosity, we provide a more predictive understanding of cathode structure evolution. The simulation framework was designed to isolate the effect of each parameter while maintaining statistical robustness through multiple independent runs. This approach ensures that the observed trends reflect true physical sensitivities rather than stochastic variability. The results show that frictional interactions, especially at CAM-SE interfaces, play a dominant role in governing both percolation and packing density. Higher CAM-SE friction impedes particle rearrangement, reducing contact area and connectivity, which in turn limits CAM utilization. Similarly, increased Young's modulus values inhibit the compression of particles, leading to elevated porosity. On the other hand, smaller SE particle sizes improve packing and percolation due to better accommodation in the voids between CAM particles. These

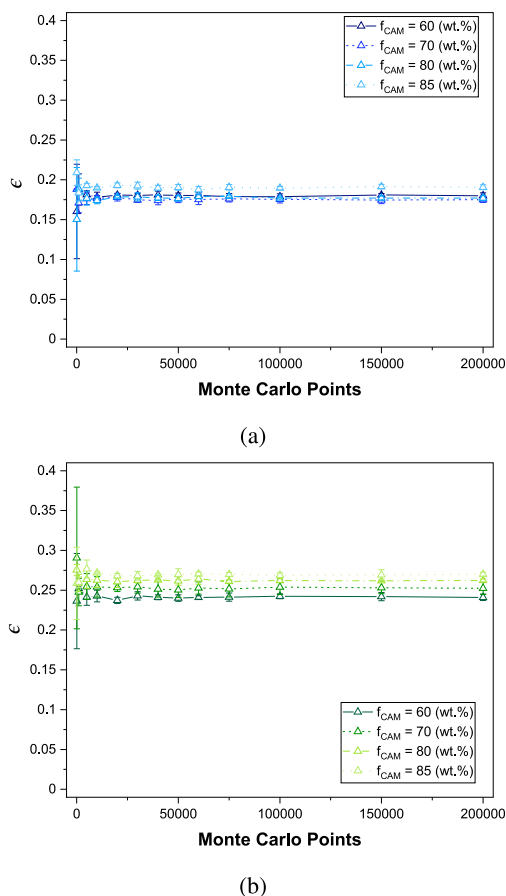


Fig. 10. Effect of Monte Carlo points for porosity calculations for (a) SE19, (b) SE45.

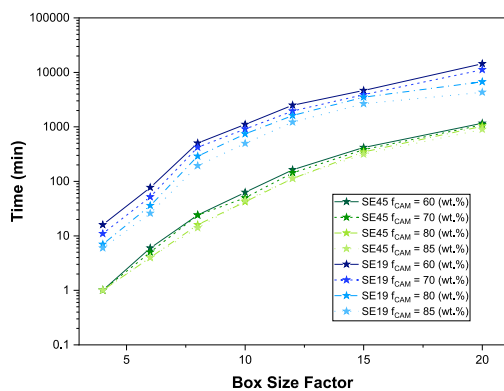


Fig. 11. Simulation time as a function of box size factor for SE19 and SE45 across different CAM loadings (f_{CAM}), highlighting the impact of SE particle size and CAM loading on computational demands.

structure-related shifts, in turn, have direct implications for electrochemical performance. Thus, understanding these parameter sensitivities not only informs model calibration but also guides experimental choices in material selection and processing conditions.

4.4. Experimental results

To determine θ_{CAM} , tests were conducted on SE45 and SE19 at pressures of 200 and 400 MPa across four f_{CAM} levels (60, 70, 80, and 85 wt% CAM). Two cells were tested per condition to ensure reproducibility, with the resulting average specific discharge capacity

Table 4

Properties of prepared DF cathode sheets.

f_{CAM} (wt.%)	SE	ϕ Thickness (μm)	ϕ Areal CAM loading ($\text{mg}_{CAM} \text{ cm}^{-2}$)
60	SE19	103	11.2
	SE45	113	12.1
70	SE19	103	13.2
	SE45	113	14.6
80	SE19	111	22.0
	SE45	112	19.6
85	SE19	106	22.5
	SE45	110	22.4

used to calculate θ_{CAM} . Table 4 summarizes the thickness and areal CAM loadings of the prepared DF cathodes.

Fig. 15 shows the charge and discharge curves for SE45 (a–d) and SE19 (e–h) under 200 and 400 MPa for different f_{CAM} . Cells at 400 MPa generally delivered higher specific discharge capacities than those at 200 MPa. For SE45, specific discharge capacities at 200 MPa were 136.5, 137.5, 104, and 50.5 mAh g^{-1} for $f_{CAM} = 60, 70, 80$, and 85 wt%, respectively, improving to 140.25 (+2.8%), 139 (+1.1%), 114 (+9.6%), and 52.5 (+4.0%) mAh g^{-1} at 400 MPa. SE19 delivered specific discharge capacities of 142.5, 144.5, 139.5, and 122.5 mAh g^{-1} at 200 MPa, which increased to 148.5 (+4.2%), 146.5 (+1.4%), 148.5 (+6.5%), and 130 (+6.1%) mAh g^{-1} at 400 MPa.

SE19 consistently outperformed SE45 across all f_{CAM} levels, particularly at higher f_{CAM} . For example, at $f_{CAM} = 85$ wt%, SE19's average specific discharge capacity was 142.6% higher at 200 MPa and 147.6% higher at 400 MPa compared to SE45. These results align with the benefits of smaller SE particles (higher λ) for maintaining high θ_{CAM} at high f_{CAM} . A reference point of 150 mAh g^{-1} was chosen for $\theta_{CAM} = 1$, representing the highest specific discharge capacity observed. Fig. 16 visualizes the comparison of average specific discharge capacities at different pressures and f_{CAM} for SE45 (a) and SE19 (b).

Fig. 17 shows the densities of cathodes pressed with SE19 and SE45 at 200 MPa and 400 MPa across different f_{CAM} levels, alongside the theoretical density, calculated using the parameters listed in Table 3(a) and the corresponding calculated porosities (b). Three key trends emerge:

- Pressure effect:** Increasing pressure from 200 to 400 MPa generally reduced ϵ . For SE19, ϵ decreased by 19.0%, 12.6%, 11.8% and 13.7% at f_{CAM} 60, 70, 80 and 85 wt% respectively. For SE45, the decreases were 22.0%, 10.9%, 7.5% and 6.1%.
- f_{CAM} effect:** Increasing f_{CAM} led to higher ϵ . For SE19, the most impactful increases occurred at 400 MPa, with ϵ rising by 9.2% from 60 to 70 wt% CAM and by 8.7% from 80 to 85 wt%. Similarly, SE45 showed the most substantial increase of 13.6% from 60 to 70 wt% at 400 MPa, with further increases of 6.7% and 6.4% observed at 200 MPa between 70 to 80 wt% and 80 to 85 wt%, respectively.
- SE particle size effect:** SE19 consistently exhibited lower ϵ compared to SE45 under the same conditions. At 200 MPa, SE19 had 23.6%, 20.5%, 12.7%, and 13.8% lower porosity than SE45 at $f_{CAM} = 60, 70, 80$, and 85 wt%. At 400 MPa, these reductions were 37.4%, 32.8%, 30.6%, and 24.9%.

These experimental results reinforce the numerical insights presented earlier by establishing clear trends between pressure, particle size, and cathode performance. Notably, the systematic comparison of SE19 and SE45 across a wide range of CAM contents and applied pressures allows us to isolate how variations in particle size distribution and compaction pressure influence both utilization and packing. The smaller SE particles in SE19 consistently enabled improved packing, as evidenced by lower porosity, and higher utilization due to better connectivity. This highlights the importance of selecting appropriate

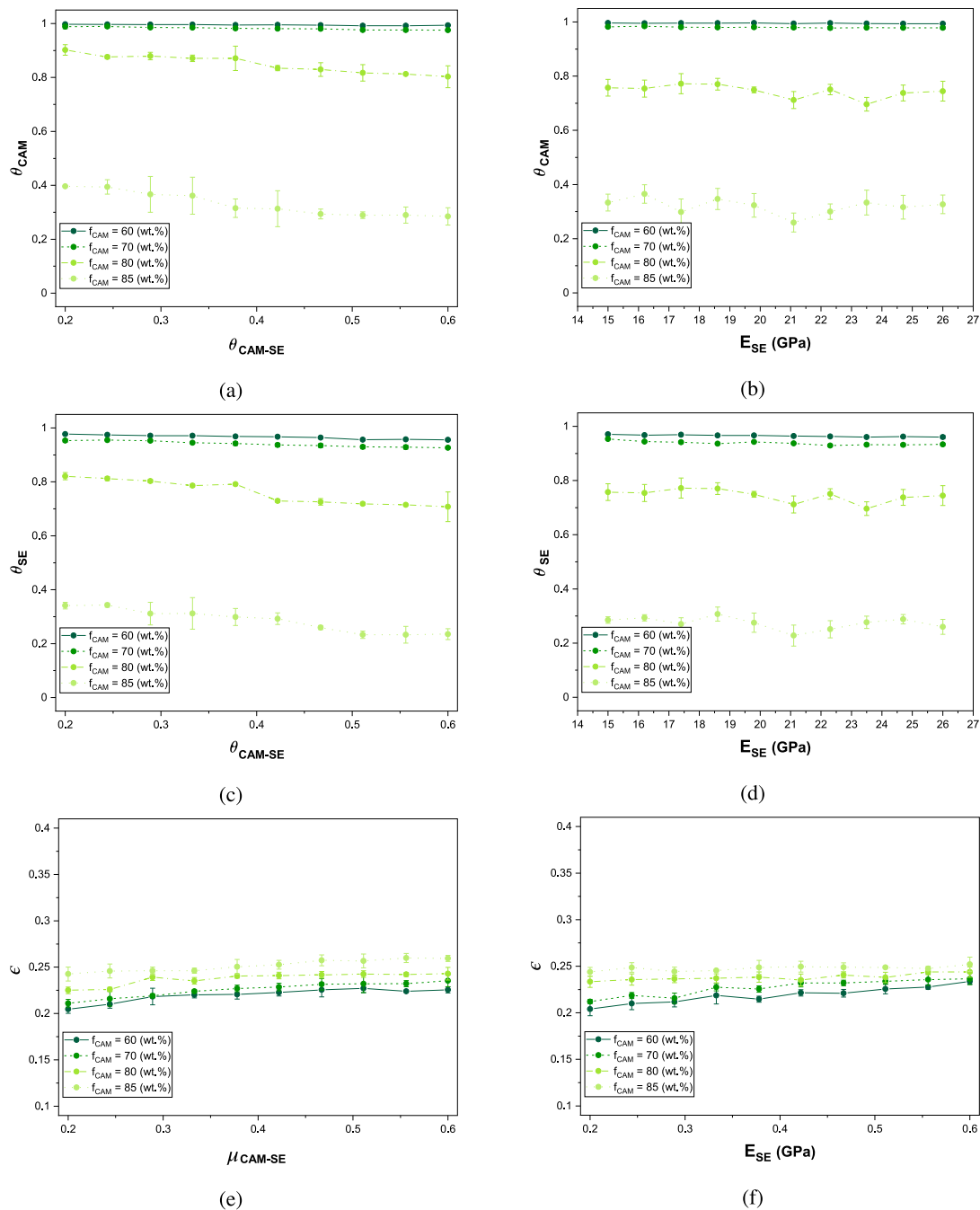


Fig. 12. Influence of CAM-SE friction coefficient ($\mu_{\text{CAM-SE}}$) on (a) CAM utilization (θ_{CAM}), (c) SE utilization (θ_{SE}), and (e) porosity (ϵ), and influence of SE Young's modulus (E_{SE}) on (b) CAM utilization, (d) SE utilization, and (f) porosity, across different CAM loadings (f_{CAM}) at 400 MPa for SE45.

SE particle sizes to ensure percolation even at higher CAM loadings. Moreover, pressure increases result in notable improvements in both utilization and porosity suggests that mechanical densification plays a crucial role in defining the electrochemical performance envelope. These observations, combined with the simulation data, emphasize how sensitive cathode performance is to microstructural design choices and underline the predictive value of the modeling framework when calibrated against experimental benchmarks.

4.5. Model comparison with experimental data

The initial assessment of the simulation model using nominal values for all eight parameters was conducted by comparing the simulation results with experimental data, focusing on θ_{CAM} and ϵ , as θ_{SE} is not

straightforward to measure experimentally. Fig. 18 provides a comprehensive comparison of the simulation and experimental data for SE19 and SE45 at 200 MPa and 400 MPa: θ_{CAM} for SE19 (a) and SE45 (b), and ϵ for SE19 (c) and SE45 (d).

Overall, the nominal parameter set shows good alignment with experimental data across the range of f_{CAM} (60, 70, 80, and 85 wt%) for both θ_{CAM} and ϵ , particularly at lower f_{CAM} (60 and 70 wt%) where the trends are closely followed under both 200 MPa and 400 MPa pressures. However, as f_{CAM} increases to 80 wt% and 85 wt%, slight discrepancies emerge. For SE19 at 80 wt% CAM, the model aligns well with the experimental data at 400 MPa, but slightly overestimates θ_{CAM} at 200 MPa. This trend continues at 85 wt% CAM, where the model slightly overestimates θ_{CAM} at 400 MPa and underestimates it at 200 MPa. For SE45, simulation results are consistent with experimental data

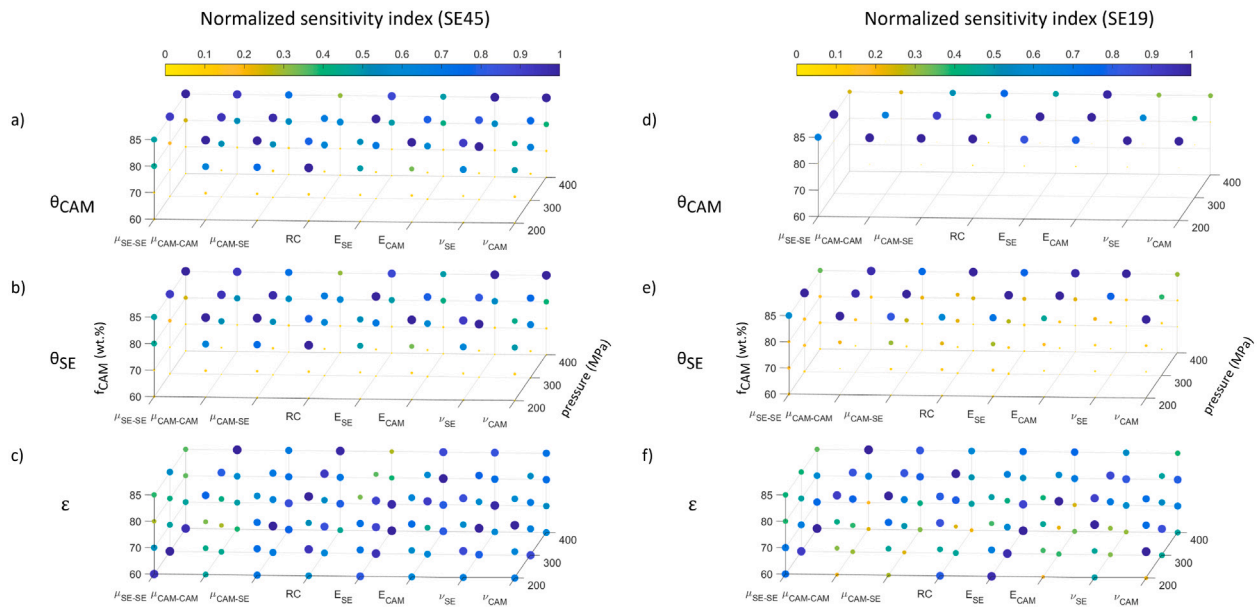


Fig. 13. Sensitivity matrices for CAM utilization (θ_{CAM}), SE utilization (θ_{SE}), and porosity (ϵ) across different CAM loadings (f_{CAM}), pressures, and SE particle sizes (SE45 and SE19), with sensitivity normalized between 0 and 1 and indicated by sphere size (smaller sphere = lower sensitivity, larger sphere = higher sensitivity): (a–c) SE45, (d–f) SE19.

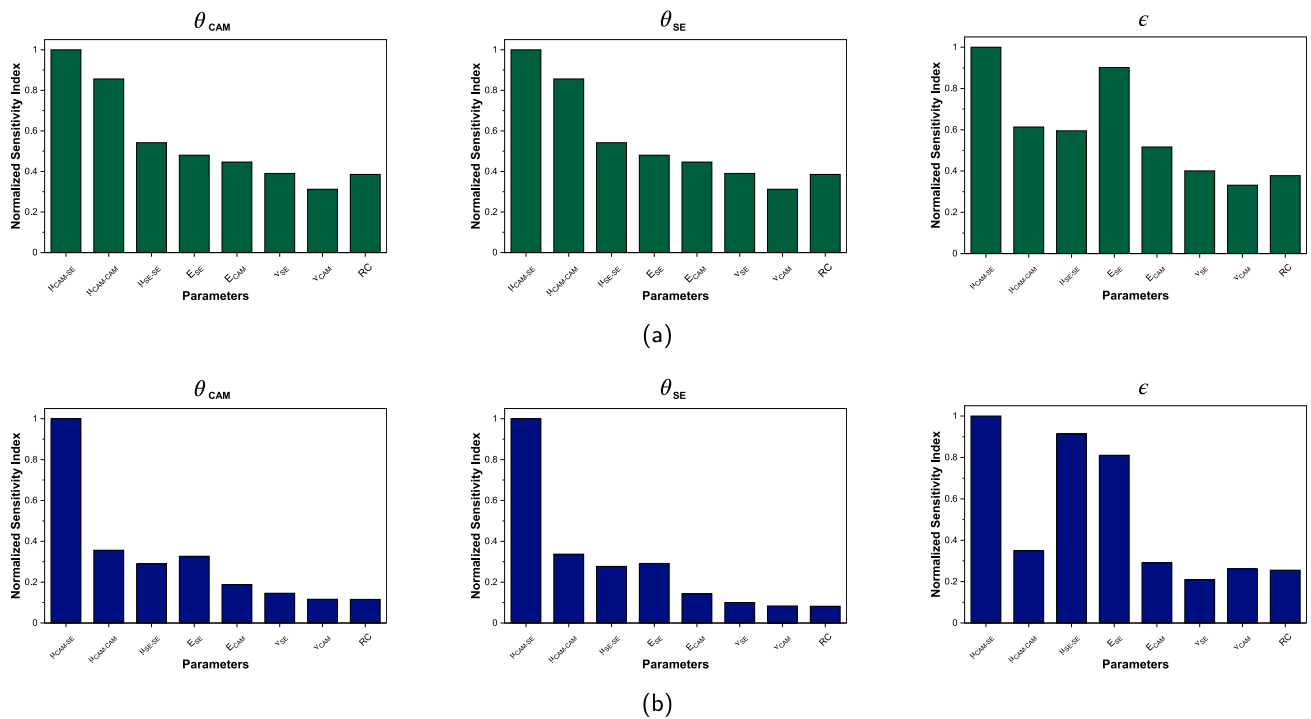


Fig. 14. Normalized sensitivity index ranking for CAM utilization (θ_{CAM}), SE utilization (θ_{SE}), and porosity (ϵ) for (a) SE45 and (b) SE19.

across all f_{CAM} , though slight underestimations occur at 85 wt% CAM. Regarding porosity ϵ , the model is relatively accurate for SE45, while consistently underestimating ϵ for SE19, particularly at higher f_{CAM} and pressures. Despite these issues, the model successfully captures the trend of increasing porosity with higher f_{CAM} , consistent with experimental behavior. These findings suggest that while the nominal parameter set provides a reliable baseline, refinement is needed for higher f_{CAM} and smaller SE particles.

Following the sensitivity analysis, friction coefficients, particularly μ_{CAM-SE} , were among the most critical parameters influencing the simulation outputs. While the Young's moduli E_{SE} and E_{CAM} also showed

sensitivity, their impact was found to be moderate in comparison to the friction coefficients. To refine the model, we unified the friction coefficients (μ_{CAM-SE} , μ_{SE-SE} , and $\mu_{CAM-CAM}$) into an effective friction coefficient (μ) across five levels (0.2, 0.3, 0.4, 0.5, 0.6) as shown in Table 5. This unification simplifies the analysis, as differentiating the coefficients individually without precise experimental measurements would be challenging. Moreover, the similar effects of the friction coefficients on simulation outcomes further justify their unification. The Young's moduli for SE and CAM, Poisson ratios, and coefficient of restitution were initially kept at nominal values to focus on the most influential parameters identified by the sensitivity analysis.

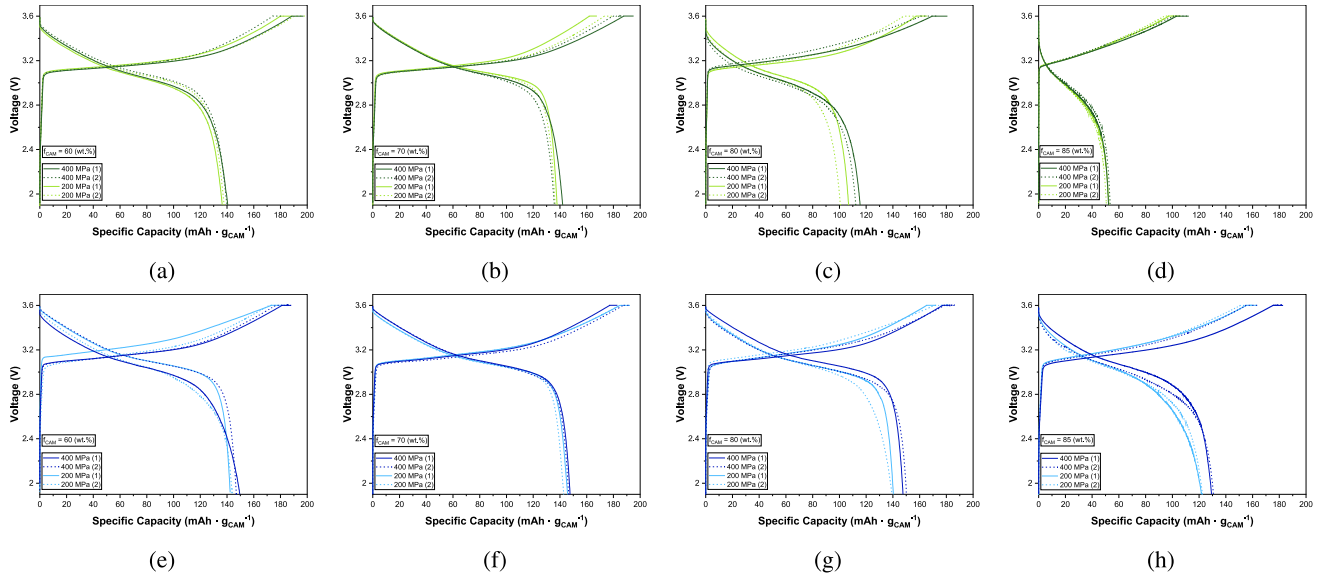


Fig. 15. First cycle charge and discharge curves at 200 MPa and 400 MPa for SE45 (a–d) and SE19 (e–h) across different CAM loadings ($f_{CAM} = 60, 70, 80$, and 85 wt%).

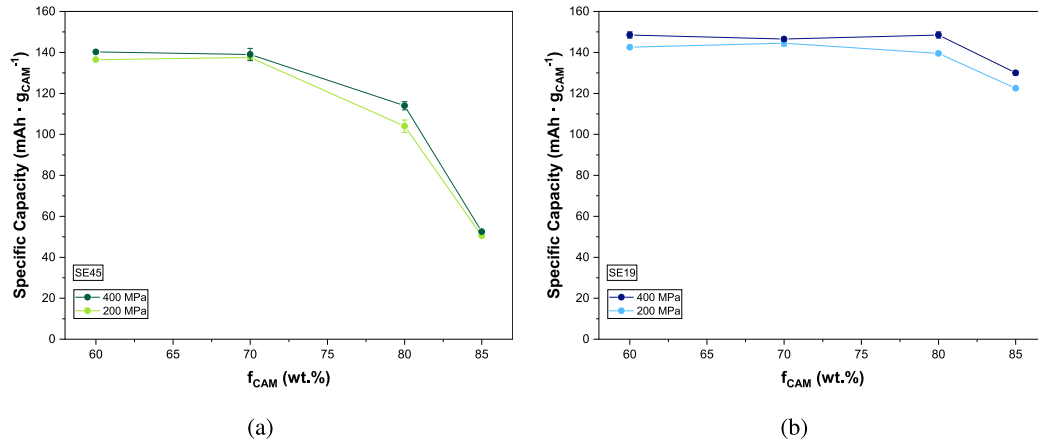


Fig. 16. Comparison of specific discharge capacities at different pressures and CAM loadings (f_{CAM}) for SE45 (a) and SE19 (b).

Table 5

Friction coefficient parameter set.

Parameter set	μ_{SE-SE}	$\mu_{CAM-CAM}$	μ_{CAM-SE}	μ
1	0.2	0.2	0.2	0.2
2	0.3	0.3	0.3	0.3
3	0.4	0.4	0.4	0.4
4	0.5	0.5	0.5	0.5
5	0.6	0.6	0.6	0.6

Fig. 19 represents a detailed analysis of the relative error in θ_{CAM} and ϵ as a function of the effective friction coefficient (μ) at various f_{CAM} under 200 and 400 MPa pressures. For θ_{CAM} , the model shows strong alignment with experimental data at lower f_{CAM} (60, 70 wt% CAM), with minimal relative error across all μ . However, as f_{CAM} increases to 80 wt% and particularly 85 wt%, the model becomes more sensitive to the variations in μ . Lower μ tend to slightly overpredict (red color) θ_{CAM} , while higher μ values shift the error towards underprediction (blue color), particularly at $f_{CAM} = 85$ wt%. A moderate friction coefficient ($\mu = 0.4$) consistently minimizes error across all scenarios, providing the most accurate model performance for both SE19 and SE45.

In contrast, the model's prediction of ϵ follows a different trend. While the error is minimal at lower f_{CAM} (60, 70 wt%), showing

a slight shift from underprediction at lower μ to near-zero or slight overprediction at higher μ , the model consistently underpredicts ϵ as f_{CAM} increases, particularly at 85 wt% CAM. Although higher μ values help slightly reduce this underprediction, the overall trend remains one of underestimation, especially for SE19 at both 200 and 400 MPa. SE45 shows relatively better alignment with experimental data. Thus, while a moderate friction coefficient ($\mu = 0.4$) offers the most balanced results, it is less effective in fully mitigating underprediction in ϵ compared to its performance with θ_{CAM} .

These results demonstrate that the simulation framework, even with nominal parameters, is capable of capturing the main trends observed experimentally, particularly at moderate CAM loadings. However, the divergence observed at high CAM fractions and for small SE particle sizes underscores the sensitivity of the model to microstructural details. The use of an effective friction coefficient proved especially useful in refining agreement with the experimental data, indicating that frictional interactions dominate behavior near the percolation threshold. Importantly, the consistency in predicted trends across both porosity and CAM utilization suggests that the underlying model captures the essential physics of particle rearrangement and compaction. By systematically aligning simulations with experimental measurements under different pressures and compositions, this study confirms that variations in particle size ratio, contact mechanics, and mechanical stiffness are

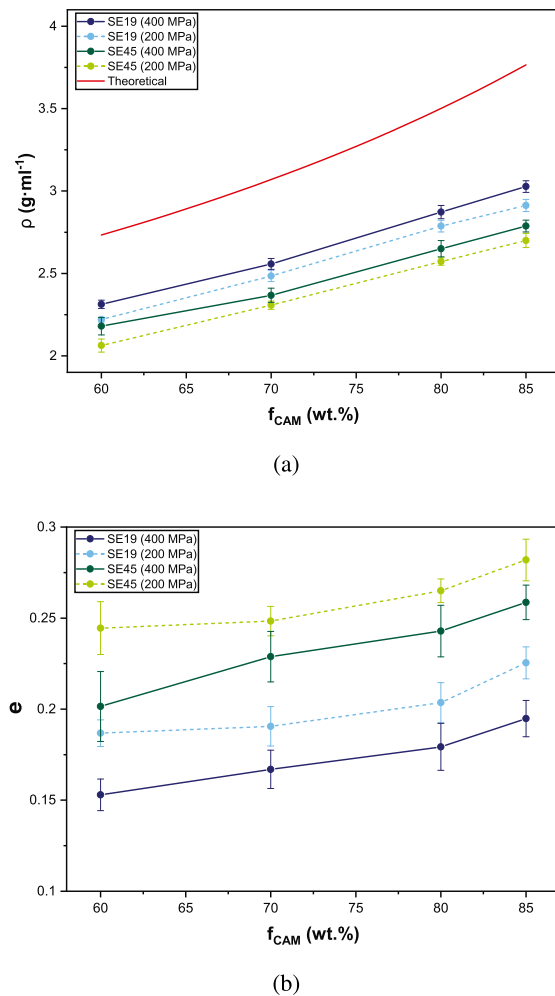


Fig. 17. Densities (a) and corresponding porosities (b) of cathodes pressed uniaxially with SE19 and SE45 across different CAM loadings (f_{CAM}) at 200 MPa and 400 MPa.

not only influential at the microscopic scale but also critically shape macroscopic performance indicators. These insights validate the use of the calibrated model as a predictive tool for guiding electrode formulation and process optimization in all-solid-state battery cathodes.

5. Discussion and outlook

5.1. Parameter sensitivity analysis

The parameter sensitivity analysis across a diverse range of simulation cases provided critical insights that will guide future research and microstructure design. The friction coefficient between CAM and SE (μ_{CAM-SE}) consistently emerged as the most sensitive parameter, emphasizing its pivotal role in influencing both percolation and porosity ϵ . This sensitivity is particularly pronounced at higher f_{CAM} , where the effectiveness of the SE percolation network diminishes as CAM content increases. In this composition window ($f_{CAM} = 60\text{--}85$ wt.%), the system forms a relatively homogeneous mixture, and the number of CAM–SE contacts statistically exceeds those of SE–SE or CAM–CAM, due to both the relative volume distributions and the particle size ratio. The smaller SE particles interact frequently with the larger CAM particles, and this is further amplified by the mechanical contrast between the two materials, specifically, the significantly higher stiffness of CAM compared to the softer SE. This difference in mechanical properties means that interfacial friction at CAM–SE contacts more directly influences how well SE particles can accommodate into the interstitial

spaces between CAM particles during compression. A high CAM–SE friction resists this accommodation and restricts densification, whereas a lower μ_{CAM-SE} enables sliding and rearrangement, leading to a more compact and connected structure. This behavior is also reflected in the increasing θ_{CAM} and decreasing ϵ observed at high λ , where smaller SE particles contribute to improved structural characteristics. This finding underscores the need for precise control and measurement of μ_{CAM-SE} to optimize cathode performance through microstructural optimization.

While other friction coefficients (μ_{SE-SE} , $\mu_{CAM-CAM}$) and Young's moduli (especially E_{SE} at lower f_{CAM}) also play significant roles, their influence is generally less pronounced compared to μ_{CAM-SE} . This suggests that frictional interactions dominate the microstructural dynamics in these systems. Although friction coefficients are not directly adjustable during electrode fabrication, the results of this study can still inform practical design decisions. Knowing that the (μ_{CAM-SE}) has the strongest influence on porosity and connectivity helps identify which material combinations and processing steps are most critical. For example, selecting CAM and SE materials with surface properties that reduce interfacial friction could improve densification during pressing. Adjusting the particle size ratio or shape may also reduce the negative effects of high friction. Additionally, processing parameters such as binder selection, mixing methods, or pressure levels can be tailored based on which interfaces are most sensitive. In this way, the sensitivity analysis provides a useful framework for guiding material selection and process optimization, even if friction itself cannot be precisely controlled. Furthermore, the insights from this work could incentivize producers regarding morphology design during synthesis, to better accommodate interfacial interactions and improve electrode performance.

The limited impact of Poisson's ratios (ν_{SE} , ν_{CAM}) and the Coefficient of Restitution (RC) across all outputs further suggests that these parameters could be deprioritized in detailed material characterizations, though they should not be entirely dismissed, as their impact could vary under different conditions.

Interestingly, SE45 displayed greater sensitivity to parameter variations compared to SE19, likely due to its lower particle size ratio (λ), making it more responsive to changes in friction and mechanical properties. This differential sensitivity between SE45 and SE19 highlights the importance of particle size distribution in determining the material's packing behavior and overall performance, particularly concerning porosity.

These findings collectively emphasize the need for targeted experimental validation, focusing on the precise measurement or calibration of the most influential parameters. This approach will ensure accurate and reliable optimization of material design for high f_{CAM} applications.

5.2. Experimental results

The experimental results demonstrate that increasing pressure from 200 MPa to 400 MPa generally enhances discharge capacity for both SE19 and SE45 across all f_{CAM} . This improvement is likely due to better particle contact and connectivity, which boosts electrochemical performance. However, while higher pressures densify the composite electrode sheets and improve performance, this must be carefully balanced, as excessively high pressures can cause pulverization and cracking in electrode materials, ultimately reducing capacity [64]. Increased pressure also decreases porosity and enhances the contact area between the CAM and SE, significantly reducing charge transfer resistance and further improving discharge capacity [65].

SE19 consistently outperformed SE45, particularly at higher f_{CAM} , due to its smaller particle size (higher λ), which improves the percolation network and reduces resistance. This aligns with findings that smaller solid electrolyte particles enhance ionic conductivity and rate performance in composite electrodes [32]. Additionally, optimizing particle size distribution, as seen with $\text{Li}_{10}\text{P}_2\text{S}_{12}$, enhances the three-dimensional balance of electron- and ion-conductive pathways, further

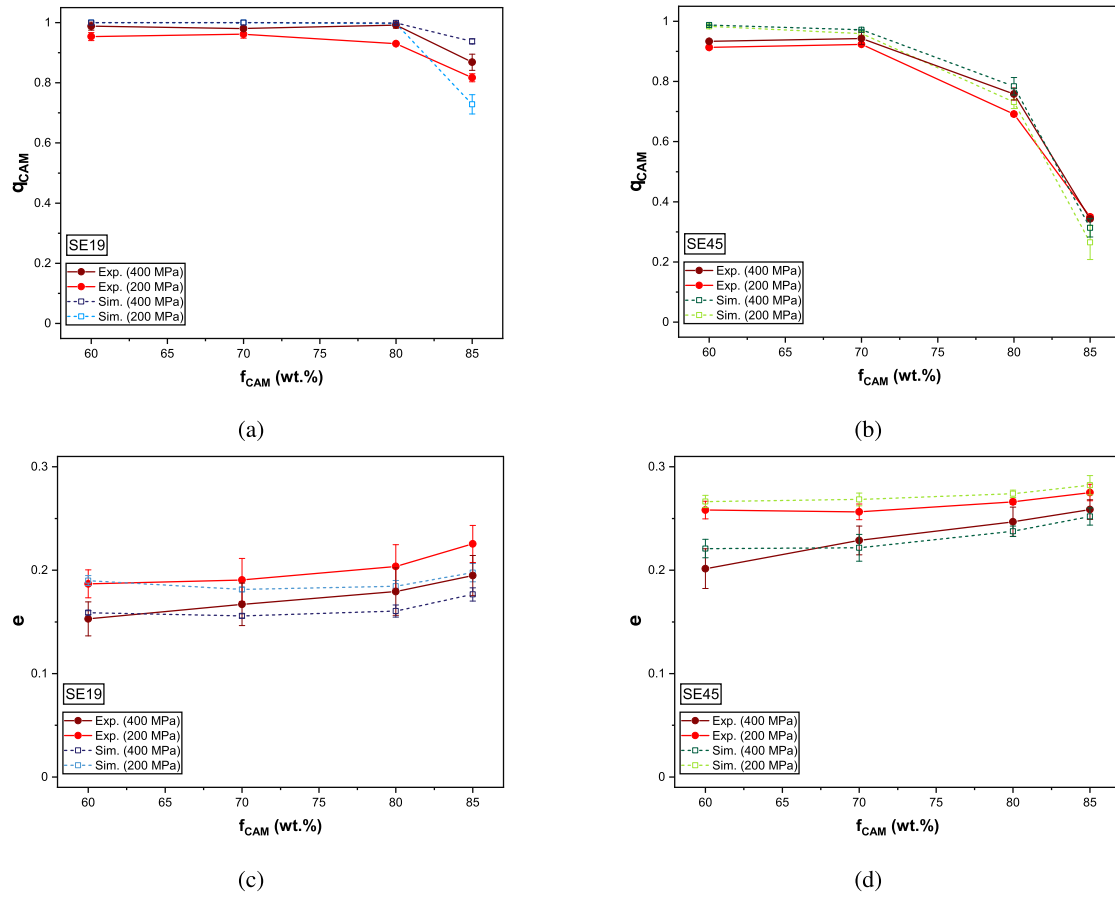


Fig. 18. Comparison of simulations and experimental data using nominal parameter values for SE19 and SE45: (a) CAM utilization for SE19, (b) CAM utilization for SE45, (c) porosity for SE19, and (d) porosity for SE45.

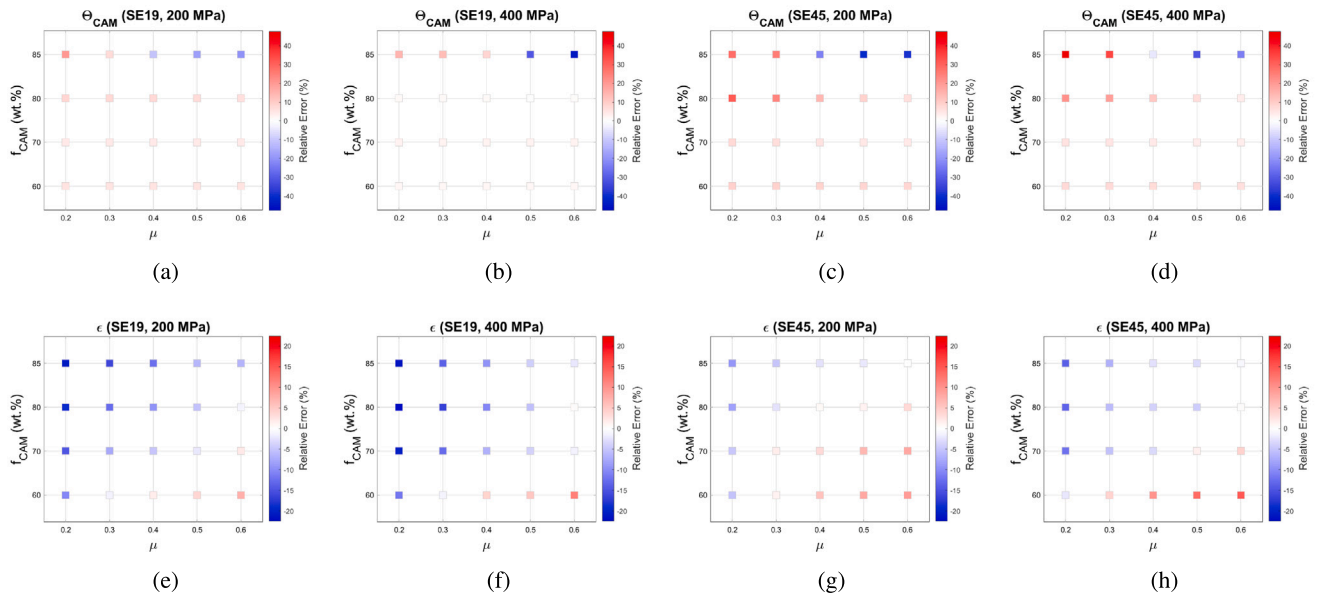


Fig. 19. Relative error analysis as a function of effective friction coefficient (μ) at various CAM loadings under 200 and 400 MPa: (a–d) CAM utilization for SE19 and SE45, (e–h) porosity for SE19 and SE45.

improving the properties of ASSBs [66]. Conversely, SE45 exhibited an earlier capacity drop at 80 wt% CAM compared to SE19's drop at 85 wt% CAM, indicating less effective particle connectivity at higher f_{CAM} . This is consistent with studies showing that high CAM fractions can reduce ionic conductivity and increase clustering, thereby lowering C-rate capability and utilization [14,15]. These findings underscore the critical role of optimizing the particle size ratio (λ) for improved cathode design.

In terms of porosity, it was shown that higher pressures reduce porosity by enhancing packing density [65], while higher f_{CAM} increases porosity, likely due to reduced SE availability to fill voids. This phenomenon is also observed in [67], where increased CAM content decreases packing efficiency and increases porosity. Furthermore, research indicates that higher CAM content, due to its significantly higher Young's modulus, can act as a stress support, decreasing the stress on SE and limiting plastic deformation. This reduces the SE's effectiveness in filling gaps, contributing to increased porosity [18]. SE19's significantly lower particle size distribution compared to SE45 suggests more effective void filling and packing of the cathode, aligning with research showing that smaller SE particles lead to better packing and reduced porosity [19].

5.3. Model assessment

The initial assessment of the simulation model highlights both its strengths and limitations. While the model shows good accuracy at lower f_{CAM} , discrepancies at higher f_{CAM} suggest that its predictive capabilities are constrained by certain assumptions and simplifications. The model's sensitivity to friction coefficients underscores the critical role of interfacial interactions in determining both θ_{CAM} and ϵ . However, the underprediction of ϵ combined with the accuracy in predicting θ_{CAM} at higher f_{CAM} and pressures indicates that the current approach may oversimplify the complexities of particle interactions and packing behavior.

To improve model accuracy, particularly under high f_{CAM} conditions, it is crucial to address factors not currently incorporated in the model. These include particle morphology and shape, which influence packing and interactions; plastic deformation, which affects the material's ability to fill voids under pressure; and mixing behavior, which can lead to non-uniform component distribution, impacting both θ_{CAM} and ϵ . Additionally, the influence of carbon additives (CA) and binder, though present in small quantities, could affect the model's predictions. Experimental studies have shown that reducing binder content to as low as 0.1 wt% does not significantly impact performance and is sufficient to obtain a dry-film cathode sheet [13]. Other studies have highlighted the negative effects of carbon black (CB), including degradation of the solid electrolyte at its interfaces [68,69]. At higher f_{CAM} , where the CAM itself acts as an electronic conductor, minimizing or removing carbon additives could further improve model accuracy.

Considering these factors, reducing binder content and carbon additives could enhance the alignment between the simulation model and experimental results. While the current model provides valuable and relatively accurate insights, further refinements that account for these adjustments could improve its accuracy, especially under conditions of high f_{CAM} . Although the model has demonstrated accuracy for both SE19 and SE45 particle sizes, further attention may be needed to address the specific requirements of different SE particle sizes. Each particle size may require distinct adjustments in the model to optimize packing and percolation behavior, indicating that a more tailored approach could yield even more precise results.

5.4. θ_{SE} behavior

Although the experimental measurement of SE utilization (θ_{SE}) was not performed in this study due to the complexity and cost, potentially requiring advanced techniques like nano-CT or FIB-SEM microstructure

reconstruction, our simulations, as shown in Fig. 20(a) and (b), offer valuable insights into θ_{SE} behavior as an indicator of cathode performance. The behavior of θ_{SE} closely parallels that of θ_{CAM} , suggesting that θ_{SE} could serve as a useful proxy for monitoring overall CAM utilization.

Notably, θ_{SE} begins to decline at lower f_{CAM} than θ_{CAM} , indicating it could be an early indicator of performance degradation in θ_{CAM} . This makes θ_{SE} particularly valuable for predicting the onset of issues within the cathode before they manifest in θ_{CAM} . Furthermore, the results demonstrate a clear dependence of θ_{SE} on pressure, highlighting that increased pressure not only improves θ_{CAM} but also enhances SE percolation and network formation. This pressure sensitivity underscores the importance of optimizing processing conditions to maintain both θ_{CAM} and θ_{SE} , especially at high f_{CAM} .

5.5. Future work and outlook

Based on the critical insights gained from analyzing key parameters influencing the DEM model's output, future work should prioritize refining and expanding these models to produce more accurate microstructures. A deeper analysis of particle size effects on utilization and porosity will be crucial for enhancing our understanding of microstructure behavior in ASSBs. Additionally, the impact of additive content, such as CA and binder, should be explored. Although these components may not need explicit simulation within DEM, they can be introduced later as part of the Carbon-Binder Domain (CBD) to evaluate the percolation behavior of electronically conducting clusters [70].

This approach will enable the use of generated microstructures not only for utilization and packing calculations but also for studying effective transport parameters, including ionic and electronic conductivity, through resistor network models [71]. This would extend the analysis beyond previous studies that focused solely on CAM and CA without considering the solid electrolyte (SE) [28]. Coupling DEM-generated microstructures with electrochemical models, as demonstrated in conventional Li-ion electrodes, could offer a promising method to quantitatively evaluate battery performance under varying conditions, such as calendaring and manufacturing processes [72].

Furthermore, assessing the effects of other processing conditions, like isostatic compression [73], on microstructure with respect to composition and particle sizes using DEM, will provide a comprehensive understanding of factors that influence ASSB electrode performance. This includes considering the stress on particles and components under these conditions [73,74]. In addition to its effects on particle stress and densification, applied pressure is known to reduce interparticle contact resistance in composite electrodes by improving mechanical contact at particle interfaces [6]. Although our current DEM framework does not explicitly model electrical or ionic transport, the observed increase in CAM connectivity and CAM utilization at higher pressures supports this trend. The reduction in porosity and enhanced structural interconnection indicate that pressure improves physical contact between particles, particularly between CAM and SE phases. This structural consolidation would be expected to reduce both electronic and ionic contact resistance in real electrodes, contributing to improved electrochemical performance. Therefore, the results presented here provide indirect but valuable insight into how applied pressure may influence not only structural but also transport-related properties in ASSB cathodes. Moreover, coupling the DEM-generated microstructures with post-processing transport models, such as resistor networks or finite element simulations, could allow direct analysis of pressure-dependent ionic and electronic conductivity based on the structural data obtained, as mentioned above.

An additional critical area of research is the effects of temperature, both in uniaxial pressing and warm isostatic compression. Warm isostatic pressing has been shown to effectively reduce porosity and improve particle contact [6], which highlights the need to explore temperature-dependent behaviors in these processes. This will require temperature-specific material data as input and consideration of how temperature affects frictional behavior and overall microstructure formation.

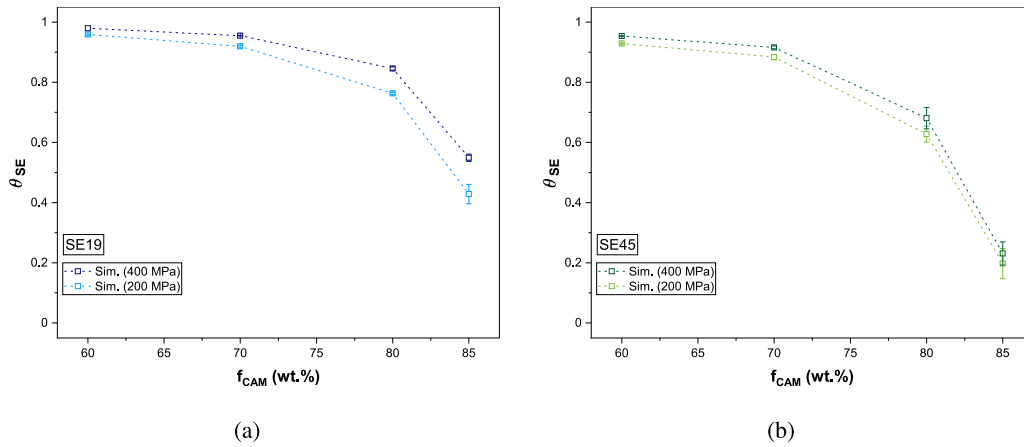


Fig. 20. SE utilization (θ_{SE}) simulation results with nominal parameter values for (a) SE19 and (b) SE45 at 200 and 400 MPa.

6. Conclusion

This study developed and evaluated a discrete element method (DEM) model to simulate the microstructural formation of solid-state battery cathodes during uniaxial cold-pressing, focusing on the effects of multiple input parameters including CAM loading (f_{CAM}), CAM and SE particle size, and pressure on key performance indicators such as CAM utilization (θ_{CAM}) and porosity (ϵ). The sensitivity analysis identified the friction coefficient between CAM and SE (μ_{CAM-SE}) as the most critical parameter influencing the model's accuracy, particularly at higher f_{CAM} . This finding underscores the importance of accurately determining interfacial friction properties to enhance the reliability of DEM models.

While the simulation results generally aligned well with experimental data at lower f_{CAM} , discrepancies emerged at higher f_{CAM} . The model tended to underestimate ϵ at smaller SE particle sizes and showed mixed accuracy in predicting θ_{CAM} , with overpredictions at lower friction coefficients and underpredictions at higher ones. Despite these challenges, the model successfully captured key trends observed experimentally, such as the increasing ϵ and decreasing θ_{CAM} with higher f_{CAM} . These findings highlight the need for further refinement of input parameters to better align the model with experimental data and improve its predictive capabilities.

The study also emphasizes the importance of incorporating additional factors into future DEM models, such as realistic particle morphology, plastic deformation, and the impact of carbon additives and binders. Including these factors could significantly enhance the model's accuracy, particularly under conditions of high f_{CAM} and varying particle sizes. Furthermore, exploring temperature-dependent behaviors and integrating advanced processing conditions, such as warm isostatic compression, could open new avenues for optimizing ASSB cathode designs.

In conclusion, this research provides a solid foundation for future work aimed at refining DEM models to optimize the design and performance of ASSB cathodes. By focusing on the precise measurement and calibration of critical parameters, as well as addressing the current model's limitations, highly accurate and reliable simulation tools can be developed that are essential for advancing ASSB technology.

CRedit authorship contribution statement

Ali Mohammad Bazzoun: Writing – original draft, Validation, Software, Methodology, Conceptualization. **Javid Piruzjam:** Writing – original draft, Software, Methodology, Conceptualization. **Steffen Hink:** Writing – review & editing, Supervision, Funding acquisition. **Lukas Rubacek:** Writing – review & editing, Supervision, Funding acquisition. **Alexander Fill:** Writing – review & editing, Supervision. **Thomas Carraro:** Writing – review & editing, Conceptualization. **Kai Peter Birke:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

The material-dependent coefficients for the force–displacement relations, as well as the viscoelastic damping factors in Eqs. (6) and (7) can respectively be written as

$$k_{ij,n} = \frac{4}{3} E^* \sqrt{R^* \delta_{ij,n}}, \quad k_{ij,t} = 8 G^* \sqrt{R^* \delta_{ij,t}} \quad (26)$$

$$\gamma_{ij,n} = -\sqrt{5} \left(\frac{\ln(RC)}{\sqrt{\ln^2(RC) + \pi^2}} \right) \sqrt{m^* k_{ij,n}},$$

$$\gamma_{ij,t} = -\sqrt{\frac{10}{3}} \left(\frac{\ln(RC)}{\sqrt{\ln^2(RC) + \pi^2}} \right) \sqrt{m^* k_{ij,t}}. \quad (27)$$

where RC is the coefficient of restitution, R^* and m^* are the equivalent radius and mass of two contacting particles with index i and j , and E^* and G^* are respectively their equivalent Young's and shear modulus, calculated as

$$\frac{1}{R^*} = \frac{1}{R_i} + \frac{1}{R_j} \quad (28)$$

$$\frac{1}{m^*} = \frac{1}{m_i} + \frac{1}{m_j} \quad (29)$$

$$\frac{1}{E^*} = \frac{1 - \nu_i^2}{E_i} + \frac{1 - \nu_j^2}{E_j} \quad (30)$$

$$\frac{1}{G^*} = \frac{2(2 - \nu_i)(1 + \nu_i)}{E_i} + \frac{2(2 - \nu_j)(1 + \nu_j)}{E_j} \quad (31)$$

where ν represents the Poisson ratio.

Similar equations of motion as described in Section 3.1 are also used for particle–wall collision, while the walls are considered to have significantly higher mass, radius and Young's modulus than the particles, hence the corresponding terms in Eqs. (28) to (31) are assumed as zero [40,75].

Truncation of the tangential force

After computing the tangential force $F_{ij,t}$ based on the spring-dashpot model, the Coulomb friction limit is enforced by truncating the tangential force if its magnitude exceeds $\mu_{ij}|F_{ij,n}|$. The final tangential force is given by:

$$F_{ij,t}^{\text{final}} = \begin{cases} F_{ij,t}, & \text{if } |F_{ij,t}| \leq \mu_{ij}|F_{ij,n}| \\ \mu_{ij}|F_{ij,n}| \frac{F_{ij,t}}{|F_{ij,t}|}, & \text{if } |F_{ij,t}| > \mu_{ij}|F_{ij,n}| \end{cases} \quad (32)$$

This truncation ensures that the tangential contact force complies with the Coulomb friction law, limiting sliding resistance to a value proportional to the normal contact force [40].

Data availability

Data will be made available on request.

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